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SHARP ENDPOINT L^p ESTIMATES FOR SCHRÖDINGER GROUPS

PENG CHEN, XUAN THINH DUONG, JI LI AND LIXIN YAN

Abstract. Let L be a non-negative self-adjoint operator acting on $L^2(X)$ where X is a space of homogeneous type with a dimension n. Suppose that the heat operator e^{-tL} satisfies the generalized Gaussian (p_0, p_0')-estimates of order m for some $1 \le p_0 < 2$. In this paper we prove *sharp* endpoint L^p-Sobolev bound for the Schrödinger group e^{itL}, that is for every $p \in (p_0, p_0')$ there exists a constant $C = C(n, p) > 0$ independent of t such that

$$\left\|(I+L)^{-s}e^{itL}f\right\|_p \le C(1+|t|)^s\|f\|_p, \quad t \in \mathbb{R}, \quad s \ge n\Big|\frac{1}{2}-\frac{1}{p}\Big|.$$

As a consequence, the above estimate holds for all $1 < p < \infty$ when the heat kernel of L satisfies a Gaussian upper bound. This extends classical results due to Feffermann and Stein, and Miyachi for the Laplacian on the Euclidean spaces $\mathbb{R}^n$. We also give an application to obtain an endpoint estimate for L^p-boundedness of the Riesz means of the solutions of the Schrödinger equations.

1. Introduction

1.1. Background. Consider the Laplace operator $\Delta = -\sum_{i=1}^n \partial_{x_i}^2$ on the Euclidean space $\mathbb{R}^n$ and the Schrödinger equation

$$\begin{cases} i\partial_t u + \Delta u = 0, \\ u|_{t=0} = f \end{cases}$$

with initial data f. Its solution can be written as

$$u(x,t) = e^{it\Delta}f(x) = \frac{1}{(2\pi)^n}\int_{\mathbb{R}^n} \widehat{f}(\xi)e^{i(\langle x,\xi\rangle + t|\xi|^2)}d\xi$$

where $\widehat{f}$ denotes the Fourier transform of f. It is well-known that the operator $e^{it\Delta}$ acts boundedly on $L^p(\mathbb{R}^n)$ only if $p = 2$; see Hörmander [22]. For $p \neq 2$, it was shown (see for example, [7, 26, 39]) that for $s > n|1/2-1/p|$, the operator $e^{it\Delta}$ maps the Sobolev space $L^p_{2s}(\mathbb{R}^n)$ into $L^p(\mathbb{R}^n)$. Equivalently, this means that $(I+\Delta)^{-s}e^{it\Delta}$ is bounded on $L^p(\mathbb{R}^n)$, and this is not the case if $0 < s < n|1/2-1/p|$. The sharp endpoint L^p-Sobolev estimate is due to Miyachi ([32, 33]), which states that for every $p \in (1,\infty)$,

$$\left\|(1+\Delta)^{-s}e^{it\Delta}f\right\|_{L^p(\mathbb{R}^n)} \le C(1+|t|)^s\|f\|_{L^p(\mathbb{R}^n)}, \quad t \in \mathbb{R}, \quad s = n\Big|\frac{1}{2}-\frac{1}{p}\Big| \tag{1.1}$$

for some positive constant $C = C(n,p)$ independent of t. The estimate (1.1) is sharp in another way: the factor $(1+|t|)^s$ can not be improved (see [32, p. 169-170]). See also Feffermann and Stein's work [19]. These results and their generalizations were in fact results on multipliers and relied

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heavily on Fourier analysis. See, for example, Ouhabaz's monograph [34, Chapter 7] for historical background and more study on the Schrödinger groups.

The purpose of this paper is to establish such sharp endpoint L^p estimate (1.1) for the operators $(e^{itL})_{t\in\mathbb{R}}$ for a large class of non-negative self-adjoint operators acting on $L^2(X)$ on a metric measure space X. Such an operator L admits a spectral resolution

$$Lf = \int_0^\infty \lambda dE_L(\lambda) f, \quad f \in L^2(X), \tag{1.2}$$

where $E_L(\lambda)$ is the projection-valued measure supported on the spectrum of L. The operator e^{itL} is defined by

$$e^{itL} f = \int_0^\infty e^{it\lambda} dE_L(\lambda) f \tag{1.3}$$

for $f \in L^2(X)$, and forms the Schrödinger group. By the spectral theorem ([31]), the operator e^{itL} is continuous on $L^2(X)$. It is interesting to investigate L^p-mapping properties for the Schrödinger group e^{itL} on $L^p(X)$ for some $p, 1 \le p \le \infty$.

As an application of our sharp endpoint L^p estimate for the Schrödinger group e^{itL}, we also aim to obtain an endpoint estimate for L^p-boundedness of the Riesz means of the solutions of the Schrödinger equations.

1.2. Assumptions and main results. Throughout the paper we assume that X is a metric space, with distance function d, and μ is a nonnegative, Borel doubling measure on X. We say that (X, d, μ) satisfies the doubling property (see Chapter 3, [11]) if there exists a constant $C > 0$ such that

$$V(x, 2r) \le CV(x, r) \quad \forall\, r > 0,\ x \in X. \tag{1.4}$$

Note that the doubling property implies the following strong homogeneity property,

$$V(x, \lambda r) \le C\lambda^n V(x, r) \tag{1.5}$$

for some $C, n > 0$ uniformly for all $\lambda \ge 1$ and $x \in X$. In Euclidean space with Lebesgue measure, the parameter n corresponds to the dimension of the space. There also exist c and $D, 0 \le D \le n$ such that

$$V(y, r) \le c\left(1 + \frac{d(x, y)}{r}\right)^D V(x, r) \tag{1.6}$$

uniformly for all $x, y \in X$ and $r > 0$. Indeed, the property (1.6) with $D = n$ is a direct consequence of triangle inequality of the metric d and the strong homogeneity property. In the cases of Euclidean spaces $\mathbb{R}^n$ and Lie groups of polynomial growth, D can be chosen to be 0.

Consider a non-negative self-adjoint operator L and numbers $m \ge 2$ and $1 \le p_0 \le 2$. We say that the semigroup e^{-tL} generated by L, satisfies the generalized Gaussian (p_0, p_0')-estimate of order m, if there exist constants $C, c > 0$ such that

$$\left\| P_{B(x,t^{1/m})} e^{-tL} P_{B(y,t^{1/m})} \right\|_{p_0 \to p_0'} \le CV(x, t^{1/m})^{-(\frac{1}{p_0} - \frac{1}{p_0'})} \exp\left(-c\left(\frac{d(x, y)^m}{t}\right)^{\frac{1}{m-1}}\right) \tag{GGE$_{p_0,p_0',m}$}$$

for every $t > 0$ and $x, y \in X$.

Note that condition ($\mathrm{GGE}_{p_0,p_0',m}$) for the special case $p_0 = 1$ is equivalent to m-th order Gaussian estimates (see for example, [6]). This means that the semigroup e^{-tL} has integral kernels $p_t(x,y)$ satisfying the following Gaussian upper estimate:

$$|p_t(x,y)| \le \frac{C}{V(x,t^{1/m})} \exp\left(-c\left(\frac{d(x,y)^m}{t}\right)^{\frac{1}{m-1}}\right) \tag{GE_m}$$

for every $t > 0, x, y \in X$, where c, C are two positive constants and $m \ge 2$. Such estimate (GE_m) is typical for elliptic or sub-elliptic differential operators of order m (see for example, [1, 2, 9, 13, 16, 17, 20, 23, 24, 34, 38, 39, 43] and the references therein). However, there are numbers of operators which satisfy generalized Gaussian estimates and, among them, there exist many for which classical Gaussian estimates (GE_m) fail. This happens, e.g., for Schrödinger operators with rough potentials [36], second order elliptic operators with rough lower order terms [28], or higher order elliptic operators with bounded measurable coefficients [14]. See also [4, 5, 6, 10, 25, 37].

Our main result is that under the generalized Gaussian estimate ($\mathrm{GGE}_{p_0,p_0',m}$) for some $1 \le p_0 < 2$, it is sufficient to ensure that such estimate (1.1) holds for the operator $(e^{itL})_{t\in\mathbb{R}}$ for $p \in (p_0, p_0')$. Our result can be stated as follows.

Theorem 1.1. *Suppose that (X,d,μ) is a space of homogeneous type with a dimension n. Suppose that L satisfies the property ($\mathrm{GGE}_{p_0,p_0',m}$) for some $1 \le p_0 < 2$. Then for every $p \in (p_0, p_0')$, there exists a constant $C = C(n,p) > 0$ independent of t such that*

$$\left\|(I+L)^{-s}e^{itL}f\right\|_p \le C(1+|t|)^s\|f\|_p, \quad t \in \mathbb{R}, \quad s \ge n\left|\frac{1}{2} - \frac{1}{p}\right|. \tag{1.7}$$

As a consequence, this estimate (1.7) holds for all $1 < p < \infty$ when the heat kernel of L satisfies a Gaussian upper bound (GE_m).

As a consequence of Theorem 1.1, we have the following result.

Corollary 1.2. *Suppose that (X,d,μ) is a homogeneous space with a dimension n. Suppose that L satisfies the property ($\mathrm{GGE}_{p_0,p_0',m}$) for some $1 \le p_0 < 2$. Then for every $p \in (p_0, p_0')$ and $s \ge n|1/2 - 1/p|$, the mapping $t \to (I+L)^{-s}e^{itL}$ is strongly continuous on $L^p(X)$.*

We now apply the result of Theorem 1.1 to study the property of the solution to the Schrödinger equation

$$\begin{cases} i\partial_t u + Lu = 0, \\ u(\cdot,0) = f. \end{cases} \tag{1.8}$$

Then we have

$$u(t,x) = e^{itL}f(x).$$

One can see that the operator e^{itL} is bounded on L^p only for $p = 2$. Following Sjöstrand [39], we define the Riesz means

$$I_s(t)(L) := st^{-s}\int_0^t (t-\lambda)^{s-1}e^{-i\lambda L}d\lambda \tag{1.9}$$

for $t > 0$, and $I_s(t)(L) = \overline{I}_s(-t)(L)$ for $t < 0$ (see also [3, 21]), and ask the question: For what values of s the operators $I_s(t)(L)$ are bounded on $L^p(X)$?

Then we have the following result.

Theorem 1.3. *Suppose that (X, d, μ) is a space of homogeneous type with a dimension n. Suppose that L satisfies the property $(\mathrm{GGE}_{\mathrm{p_0,p_0',m}})$ for some $1 \le p_0 < 2$. Then for every $p \in (p_0, p_0')$, there exists a constant $C = C(n, p) > 0$ independent of t such that*

$$\|I_s(t)(L)f\|_p \le C\|f\|_p, \quad t \in \mathbb{R}\backslash\{0\}, \quad s \ge n\Big|\frac{1}{2} - \frac{1}{p}\Big|. \tag{1.10}$$

As a consequence, this estimate (1.10) holds for all $1 < p < \infty$ when the heat kernel of L satisfies a Gaussian upper bound (GE_m).

It is known that such estimate (1.10) holds due to Sjöstrand [39] for the Laplacian $-\Delta$ on $\mathbb{R}^n$([39]); see also Thangavelu's work [42] for the harmonic oscillator $-\Delta + |x|^2$ on $\mathbb{R}^n$.

The proof of Theorem 1.1 and Corollary 1.2 will be given in Section 3. The proof of Theorem 1.3 will be given in Section 4.

1.3. Comments on the results and methods of the proof. On Lie groups with polynomial growth and manifolds with non-negative Ricci curvature, similar results as in (1.1) for $s > n|1/2 - 1/p|$ have been first announced by Lohoué in [29], then Alexopoulos obtained them in [1]. There, the method is to replace Fourier analysis by the finite propagation speed of the associated wave equation [41]. In the abstract setting of operators on metric measure spaces, Carron, Coulhon and Ouhabaz [9] showed L^p-boundedness of suitable regularizations of the Schrödinger group e^{itL} provided L satisfies Gaussian estimate (GE_m). They proposed a different approach to use some techniques introduced by Davies [13]: the Gaussian semigroup estimates can be extended from real times $t > 0$ to complex times $z \in \mathbb{C}^+ = \{z \in \mathbb{C} : \mathrm{Re}\, z > 0\}$ such that

$$\|e^{-zL}\|_{p\to p} \le C\left(\frac{|z|}{\mathrm{Re}\, z}\right)^{n|\frac{1}{2}-\frac{1}{p}|+\epsilon}, \quad \forall z \in \mathbb{C}^+. \tag{1.11}$$

On the other hand, for every $f \in L^2 \cap L^p$ and $s \ge 0$,

$$(I + L)^{-s}e^{itL}f = \frac{1}{\Gamma(s)}\int_0^\infty e^{-u}u^{s-1}e^{-(u-it)L}f\,du,$$

where Γ is the Euler Gamma function. From (1.11), we see that for $s > n|1/2 - 1/p|$,

$$\|(I + L)^{-s}e^{itL}\|_{p\to p} \le C\int_0^\infty e^{-u}u^{s-1}\left(\sqrt{\frac{u^2 + t^2}{u^2}}\right)^{n|\frac{1}{2}-\frac{1}{p}|+\epsilon} du \tag{1.12}$$

and so (1.7) holds for $s > n|1/2 - 1/p|$. The Gaussian bound (GE_m) assumption on L was further weakened to the generalized Gaussian estimates $(\mathrm{GGE}_{\mathrm{p_0,p_0',m}})$ by Blunck [4, Theorem 1.1] where the estimate (1.11) was improved to get $\epsilon = 0$, i.e.

$$\|e^{-zL}\|_{p\to p} \le C\left(\frac{|z|}{\mathrm{Re}\, z}\right)^{n|\frac{1}{2}-\frac{1}{p}|}, \quad \forall z \in \mathbb{C}^+ \tag{1.13}$$

for all $p \in [p_0, p_0']$ with $p \neq \infty$, and so (1.7) holds for $s > n|1/2 - 1/p|$. However, it is direct to see that the integral in (1.12) is ∞ when $s = n|1/2 - 1/p|$.

It was an open question whether estimate (1.7) holds with $s = n|1/2 - 1/p|$. Based on estimate (1.13), it is straightforward to obtain sharp L^p frequency truncated estimates for e^{itL} that for every $p \in (p_0, p_0')$ and $k \in \mathbb{Z}^+$,

$$\|e^{itL}\phi(2^{-k}L)f\|_p \leq C(1 + 2^k|t|)^s\|f\|_p, \quad t \in \mathbb{R}, \quad s = n\Big|\frac{1}{2} - \frac{1}{p}\Big| \tag{1.14}$$

uniformly for ϕ in bounded subsets of $C_0^\infty(\mathbb{R})$, by writing

$$e^{itL}\phi(2^{-k}L)f = e^{-(2^{-k}-it)L}[\phi_e(2^{-k}L)](f)$$

where $\phi_e(\lambda) = e^\lambda\phi(\lambda)$ and then applying (1.13) to $e^{-(2^{-k}-it)L}$ and [5, Theorem 1.1] to $\phi_e(2^{-k}L)$, respectively (for more details, see Proposition 3.1 below). As a consequence of (1.14), it follows by a standard scaling argument ([23, p. 193]) that for every $p \in (p_0, p_0')$ and for every $\epsilon > 0$,

$$\|(I + L)^{-s-\epsilon}e^{itL}f\|_p \leq C(1 + |t|)^s\|f\|_p, \quad t \in \mathbb{R}, \quad s = n\Big|\frac{1}{2} - \frac{1}{p}\Big|. \tag{1.15}$$

We would like to mention that in [12], D'Ancona and Nicola used a commutator argument and a reduction to amalgam spaces and followed the methods of Jensen-Nakamura [23, 24] to obtain estimates (1.14) and (1.15) for the Schrödinger group e^{itL} for $p \in [p_0, p_0']$ in the Euclidean spaces $\mathbb{R}^n$. However, as in [12, p.1021], the authors remarked that "*Another interesting issue is the validity of* (1.15) *with* $\epsilon = 0$. *Indeed, for* $L = -\Delta$ *in* $\mathbb{R}^n$ *and* $1 < p < \infty$, *the estimate* (1.15) *was proved with* $\epsilon = 0$ *(and* $t = 1$*) in* [33]*, but this sharp form seems out of reach in the present generality, even for fixed* t*"*. Under an additional condition which is the operator e^{itL} being bounded in suitable modulation spaces (see [12, Section 5] for the definition), it was proved in [12] that estimate (1.15) holds with $\epsilon = 0$ in the setting of $\mathbb{R}^n$. See also previous related results [8, 23, 24].

Our main result, Theorem 1.1, gives the sharp endpoint estimate (1.15) for the Schrödinger group e^{itL} with $\epsilon = 0$, namely with the optimal number of derivatives and the optimal time growth for the factor $(1 + |t|)^s$ in (1.15). The proof of Theorem 1.1 is different from those of Fefferman and Stein [19] and Miyachi [32, 33] where the results rely heavily on Fourier analysis. In our setting, we do not have Fourier transform at our disposal. We also do not assume that the heat kernel $p_t(x, y)$ satisfies the standard regularity condition, thus standard techniques of Calderón–Zygmund theory ([40]) are not applicable. The lack of smoothness of the kernel will be overcome in Proposition 2.3 below by using some off-diagonal estimates on heat semigroup of non-negative self-adjoint operators, and some techniques in the theory of singular integrals with rough kernels, which lies beyond the scope of the standard Calderón-Zygmund theory (see for example, [2, 5, 6, 10, 16, 17, 18, 25, 34, 37] and the references therein). More specifically, by duality we are reduced to prove the estimate for $2 < p < p_0'$, which will follow by the Littlewood-Paley inequality and a variant of the Fefferman-Stein sharp function (see [2, 18, 19, 30, 37]),

$$\|e^{itL}f\|_p \leq C\||T_\varphi f\|_p \leq C\|\mathfrak{M}_2(|T_\varphi f|)\|_p \leq C_p(\|\mathfrak{M}^\#_{T_\varphi,L,K}f\|_p + \|f\|_p), \tag{1.16}$$

where

$$T_\varphi f(x) = \left(\sum_{k\geq 0} |\varphi_k(L)e^{itL}f(x)|^2\right)^{1/2} \tag{1.17}$$

for some cut-off function $\varphi \in C_0^\infty([1/2,2])$, where $\varphi_k(\lambda) = \varphi(2^{-k}\lambda)$, $k \geq 1$ and $\varphi_0(\lambda)+\sum_{k\geq 1}\varphi_k(\lambda) \equiv 1$ for $\lambda > 0$, and for a large $K \in \mathbb{N}$,

$$\mathfrak{M}^{\#}_{T_\varphi,L,K}f(x) \quad = \quad \sup_{B\ni x}\left(\fint_B \left|T(I-e^{-r_B^m L})^K f(y)\right|^2 d\mu(y)\right)^{1/2}. \tag{1.18}$$

We then use a variant of an argument in [27, 35] to decompose the function $\mathfrak{M}^{\#}_{T_\varphi,L,K}f$ into several components so that we can employ the off-diagonal estimates (1.20) below. Then we show that the function $\mathfrak{M}^{\#}_{T_\varphi,L,K}f$ is in L^p by using estimate (1.14) for the Schrödinger group e^{itL}. We note that in the case that L is the Laplace operator Δ on $\mathbb{R}^n$, the kernel estimate relies heavily on Fourier analysis since the operator $e^{it\Delta}\varphi(2^{-k}\Delta)$ has the convolution kernel

$$K_{e^{it\Delta}\varphi(2^{-k}\Delta)}(x) = \frac{2^{kn/2}}{(2\pi)^n}\int_{\mathbb{R}^n}\varphi(|\xi|^2)e^{i(2^{k/2}\langle x,\,\xi\rangle+2^k t|\xi|^2)}d\xi,$$

one then uses integration by parts to obtain that for every $M > 0$,

$$\left|K_{e^{it\Delta}\varphi(2^{-k}\Delta)}(x)\right| \leq C2^{kn/2}(1+2^{k/2}|x|)^{-M} \tag{1.19}$$

whenever $|x| \geq 2^{k/2+4}$ and $t \in [0,1]$ (see for example, [35, page 62]). However, when L is a general non-negative self-adjoint operator acting on the space $L^2(X)$ satisfying ($\mathrm{GGE}_{\mathrm{p_0,p_0',m}}$) with $p_0 \in [1,2)$, such estimate (1.19) may or may not hold. In our setting, we need the following off-diagonal estimate of the operator $e^{itL}\varphi(2^{-k}L)$ (see Proposition 2.3 below): For every $M > 0$, there exists a positive constant $C = C(n,m,M)$ independent of t such that

$$\left\|P_{B_1}e^{itL}\varphi(2^{-k}L)P_{B_2}f\right\|_2 \leq C\left(1+\frac{d(B_1,B_2)}{2^{(m-1)k/m}(1+|t|)}\right)^{-M}\|P_{B_2}f\|_2, \quad t\in\mathbb{R} \tag{1.20}$$

for all balls $B_1, B_2 \subset X$ with radius $r_{B_1} = r_{B_2} \geq c2^{(m-1)k/m}(1+|t|)$ for some $c \geq 1/4$, and $d(B_1,B_2) \geq 6r_{B_1}$. This new estimate is crucial for the proof of Theorem 1.1.

The paper is organized as follows. In Section 2 we provide some preliminary results on off-diagonal estimates of the operator $e^{itL}\varphi(2^{-k}L)$ and spectral multipliers and Littlewood-Paley theory, which we need later, mainly to prove (1.20) in Proposition 2.3. The proof of Theorem 1.1 will be given in Section 3. In Section 4 we will apply Theorem 1.1 to obtain L^p-boundedness of the Riesz means of the solution to the Schrödinger equation.

List of notations.

- (X,d,μ) denotes a metric measure space with a distance d and a measure μ.
- L is a non-negative self-adjoint operator acting on the space $L^2(X)$.
- For $x \in X$ and $r > 0$, $B(x,r) = \{y \in X : d(x,y) < r\}$ and $V(x,r) = \mu(B(x,r))$.
- For $B = B(x_B, r_B)$, $A(x_B, r_B, 0) = B$ and $A(x_B, r_B, j) = B(x_B,(j+1)r_B)\backslash B(x_B, jr_B)$ for $j = 1,2,\ldots$.
- $\delta_R F$ is defined by $\delta_R F(x) = F(Rx)$ for $R > 0$ and Borel function F supported on $[-R,R]$.
- $\lfloor t\rfloor$ denotes the integer part of t for any positive real number t.
- $\mathbb{N}$ is the set of positive integers.

- For $p \in [1,\infty]$, $p' = p/(p-1)$.
- For $1 \le p \le \infty$ and $f \in L^p(X, \mathrm{d}\mu)$, $\|f\|_p = \|f\|_{L^p(X,\mathrm{d}\mu)}$.
- $\langle \cdot, \cdot \rangle$ denotes the scalar product of $L^2(X, \mathrm{d}\mu)$.
- For $1 \le p, q \le +\infty$, $\|T\|_{p\to q}$ denotes the operator norm of T from $L^p(X, \mathrm{d}\mu)$ to $L^q(X, \mathrm{d}\mu)$.
- If T is given by $Tf(x) = \int K(x,y) f(y) d\mu(y)$, we denote by K_T the kernel of T.
- Given a subset $E \subseteq X$, χ_E denotes the characteristic function of E and $P_E f(x) = \chi_E(x) f(x)$.
- For every $B \subset X$, we write $\fint_B f d\mu(y) = \mu(B)^{-1} \int_B f(y) d\mu(y)$.
- For $1 \le r < \infty$, $\mathfrak{M}_r$ denotes the uncentered r-th maximal operator over balls in X, that is

$$\mathfrak{M}_r f(x) = \sup_{B \ni x} \left(\fint_B |f(y)|^r d\mu(y) \right)^{1/r}.$$

For simplicity we denote by $\mathfrak{M}$ the Hardy-Littlewood maximal function $\mathfrak{M}_1$.

2. Off-diagonal estimates and spectral multipliers

In this section we assume that (X, d, μ) is a space of homogeneous type with a dimension n in (1.5) and that L is a self-adjoint non-negative operator in $L^2(X)$ satisfying the generalized Gaussian estimate ($\mathrm{GGE}_{p_0,p_0',\mathrm{m}}$) for some $1 \le p_0 < 2$.

2.1. Off-diagonal estimates. We start by collecting some properties of the generalized Gaussian estimates obtained by Blunck and Kunstmann, see for example, [4, 5, 6, 25] and the references therein. For every $j \ge 1$, we recall that $A(x_B, r_B, j) = B(x_B, (j+1)r_B) \backslash B(x_B, j r_B)$. The following result originally stated in [25, Lemma 2.5] (see also [4, Theorem 2.1]) shows that generalized Gaussian estimates can be extended from real times $t > 0$ to complex times $z \in \mathbb{C}$ with $\mathrm{Re} z > 0$. Recall that χ_E denotes the characteristic function of $E \subseteq X$ and set $P_E f(x) = \chi_E(x) f(x)$.

Lemma 2.1. *Let $m \ge 2$ and $1 \le p \le 2 \le q \le \infty$, and L be a non-negative self-adjoint operator on $L^2(X)$. Assume that there exist constants $C, c > 0$ such that for all $t > 0$, and all $x, y \in X$,*

$$\left\| P_{B(x,t^{1/m})} e^{-tL} P_{B(y,t^{1/m})} \right\|_{p\to q} \le C V(x, t^{1/m})^{-(\frac{1}{p}-\frac{1}{q})} \exp\Big(- c \Big(\frac{d(x,y)}{t^{1/m}} \Big)^{\frac{m}{m-1}} \Big).$$

Let $r_z = (\mathrm{Re}\, z)^{\frac{1}{m}-1} |z|$ for each $z \in \mathbb{C}$ with $\mathrm{Re}\, z > 0$.

(i) *There exist two positive constants C' and c' such that for all $r > 0$, $x \in X$, and $z \in \mathbb{C}$ with $\mathrm{Re}\, z > 0$,*

$$\begin{aligned} &\left\| P_{B(x,r)} e^{-zL} P_{B(y,r)} \right\|_{p\to q} \\ &\quad \le \; C' V(x,r)^{-(\frac{1}{p}-\frac{1}{q})} \Big(1 + \frac{r}{r_z} \Big)^{n(\frac{1}{p}-\frac{1}{q})} \Big(\frac{|z|}{\mathrm{Re}\, z} \Big)^{n(\frac{1}{p}-\frac{1}{q})} \exp\Big(- c' \Big(\frac{d(x,y)}{r_z} \Big)^{\frac{m}{m-1}} \Big). \end{aligned}$$

(ii) *There exist two positive constants C'' and c'' such that for all $r > 0$, $x \in X$, $k \in \mathbb{N}$ and $z \in \mathbb{C}$ with $\mathrm{Re}\, z > 0$,*

$$\begin{aligned} &\left\| P_{B(x,r)} e^{-zL} P_{A(x,r,k)} \right\|_{p\to q} \\ &\quad \le \; C'' V(x,r)^{-(\frac{1}{p}-\frac{1}{q})} \Big(1 + \frac{r}{r_z} \Big)^{n(\frac{1}{p}-\frac{1}{q})} \Big(\frac{|z|}{\mathrm{Re}\, z} \Big)^{n(\frac{1}{p}-\frac{1}{q})} k^n \exp\Big(- c'' \Big(\frac{r}{r_z} k \Big)^{\frac{m}{m-1}} \Big). \end{aligned}$$

Proof. For the detailed proof we refer readers to [25]. Here we only mention that the proof of Lemma 2.1 relies on the Phragmén-Lindelöf theorem. □

Next suppose that $m \geq 2$. We say that the semigroup e^{-tL} generated by non-negative self-adjoint operator L satisfies *m-th order Davies-Gaffney estimates*, if there exist constants $C, c > 0$ such that for all $t > 0$, and all $x, y \in X$,

$$\left\|P_{B(x,t^{1/m})}e^{-tL}P_{B(y,t^{1/m})}\right\|_{2\to 2} \leq C\exp\left(-c\left(\frac{d(x,y)}{t^{1/m}}\right)^{\frac{m}{m-1}}\right). \tag{DG$_m$}$$

Note that if condition (GGE$_{p_0,p_0',m}$) holds for some $1 \leq p_0 \leq 2$ with $p_0 < 2$, then the semigroup e^{-tL} satisfies estimate (DG$_m$).

The following Lemma describes a useful consequence of m-order Davies-Gaffney estimates (see [37, Lemma 2.2]).

Lemma 2.2. *Let $m \geq 2$ and L satisfies the Davies-Gaffney estimates (DG$_m$). Then for every $M > 0$, there exists a constant $C = C(M)$ such that for every $j = 2, 3, \ldots$*

$$\left\|P_B F(L) P_{A(x_B,r_B,j)}\right\|_{2\to 2} \leq C j^{-M}(\sqrt[m]{R}r_B)^{-(M+n)}\|\delta_R F\|_{W_2^{M+n+1}} \tag{2.1}$$

for all balls $B \subseteq X$, and all Borel functions F such that supp $F \subseteq [-R, R]$.

Proof. Let $G(\lambda) = (\delta_R F)(\lambda)e^{\lambda}$. In virtue of the Fourier inversion formula

$$G(L/R)e^{-L/R} = \frac{1}{2\pi}\int_{\mathbb{R}} e^{(i\tau-1)R^{-1}L}\hat{G}(\tau)d\tau$$

so

$$\|P_B F(L) P_{A(x_B,r_B,j)}\|_{2\to 2} \leq \frac{1}{2\pi}\int_{\mathbb{R}} |\hat{G}(\tau)|\,\|P_B e^{(i\tau-1)R^{-1}L}P_{A(x_B,r_B,j)}\|_{2\to 2}d\tau.$$

By (ii) of Lemma 2.1 (with $r_z = \sqrt{1+\tau^2}/\sqrt[m]{R}$),

$$\begin{aligned}
\|P_B e^{(i\tau-1)R^{-1}L}P_{A(x_B,r_B,j)}\|_{2\to 2} &\leq C j^n \exp\left(-c\left(\frac{\sqrt[m]{R}jr_B}{\sqrt{1+\tau^2}}\right)^{\frac{m}{m-1}}\right)\\
&\leq C_M j^n\left(\frac{\sqrt[m]{R}jr_B}{\sqrt{1+\tau^2}}\right)^{-M-n}\\
&\leq C j^{-M}(1+\tau^2)^{\frac{M+n}{2}}(\sqrt[m]{R}r_B)^{-(M+n)}.
\end{aligned}$$

Therefore (compare [17, (4.4)])

$$\begin{aligned}
&\|P_B F(L) P_{A(x_B,r_B,j)}\|_{2\to 2}\\
&\leq C j^{-M}(\sqrt[m]{R}r_B)^{-(M+n)}\int_{\mathbb{R}} |\hat{G}(\tau)|(1+\tau^2)^{\frac{M+n}{2}}d\tau\\
&\leq C j^{-M}(\sqrt[m]{R}r_B)^{-(M+n)}\left(\int_{\mathbb{R}} |\hat{G}(\tau)|^2(1+\tau^2)^{M+n+1}d\tau\right)^{1/2}\left(\int_{\mathbb{R}}(1+\tau^2)^{-1}d\tau\right)^{1/2}\\
&\leq C j^{-M}(\sqrt[m]{R}r_B)^{-(M+n)}\|G\|_{W_2^{M+n+1}}.
\end{aligned}$$

However, supp $F \subseteq [-R, R]$ and supp $\delta_R F \subseteq [-1, 1]$ so

$$\|G\|_{W_2^{M+n+1}} \leq C\|\delta_R F\|_{W_2^{M+n+1}}.$$

This completes the proof of Lemma 2.2. □

The proof of Theorem 1.1 relies on the following off-diagonal estimates for $e^{itL}\phi_k(L)$, where $\phi \in C_0^\infty([1/4, 4])$ is a cut-off function and $\phi_k(s) = \phi(2^{-k}s)$ for every $k \geq 1$.

Proposition 2.3. *Let $m \geq 2$ and L satisfies the Davies-Gaffney estimates (DG_m). For every $M > 0$, $K \geq 1$, $s > 0$, $t \in \mathbb{R}$ and $k \geq 1$, there exists a constant $C = C(M, n, K)$ independent of t, s, and k such that*

$$\left\|P_{B_1}(I - e^{-sL})^K e^{itL}\phi_k(L)P_{B_2}f\right\|_2 \leq C\left(1 + \frac{d(B_1, B_2)}{2^{(m-1)k/m}(1 + |t|)}\right)^{-M} \|P_{B_2}f\|_2 \tag{2.2}$$

for all $B_i \subset X$ with $r_{B_1} = r_{B_2} \geq c2^{(m-1)k/m}(1 + |t|)$ for some $c \geq 1/4$, and $d(B_1, B_2) \geq 6r_{B_1}$.

To prove Proposition 2.3, we need the following Lemmas 2.4 and 2.5.

Lemma 2.4. *Let $m \geq 2$ and L satisfies the Davies-Gaffney estimates (DG_m). Then for every $M > 0$, $k \in \mathbb{N}^+$ and $t \in \mathbb{R}$, there exists a constant $C = C(M, m, n)$ independent of t and k such that for every $j = 2, 3, \ldots$*

$$\left\|P_B e^{-(2^{-k}-it)L}P_{A(x_B,r_B,j)}f\right\|_2 \leq Cj^{-M}\left(1 + \frac{r_B}{2^{(m-1)k/m}(1 + |t|)}\right)^{-M} \|P_{A(x_B,r_B,j)}f\|_2 \tag{2.3}$$

for all balls $B \subset X$ with $r_B \geq c2^{(m-1)k/m}(1 + |t|)$ for some $c \geq 1/4$.

As a consequence, we have

$$\left\|P_B e^{-(2^{-k}-it)L}P_{X\setminus 2B}f\right\|_2 \leq C\mu(B)^{1/2}\mathfrak{M}_2(f)(x)$$

for all balls $B \subset X$ with $r_B \geq c2^{(m-1)k/m}(1 + |t|)$ for some $c > 1/4$ and for every $x \in B$.

Proof. Note that

$$\|P_B e^{-(2^{-k}-it)L}P_{X\setminus 2B}f\|_2 \leq \sum_{j=2}^{\infty} \|P_B e^{-zL}P_{A(x_B,r_B,j)}f\|_2$$

with $z = (2^{-k} - it)$. It is clear that $\operatorname{Re} z = 2^{-k} > 0$, and so $r_z = (\operatorname{Re} z)^{\frac{1}{m}-1}|z| = 2^{(m-1)k/m}\sqrt{|t|^2 + 2^{-2k}}$. By (ii) of Lemma 2.1, we see that for every ball $B \subset X$ with $r_B \geq 2^{(m-1)k/m}(1 + |t|)$, $k \geq 0$,

$$\begin{aligned}\left\|P_B e^{-(2^{-k}-it)L}P_{A(x_B,r_B,j)}\right\|_{2\to 2} &\leq Cj^n \exp\left(-c\left(\frac{r_B j}{2^{(m-1)k/m}\sqrt{2^{-2k} + |t|^2}}\right)^{\frac{m}{m-1}}\right) \\ &\leq C_M j^{-M+n}\left(1 + \frac{r_B}{2^{(m-1)k/m}(1 + |t|)}\right)^{-M} \qquad (2.4)\end{aligned}$$

for every $M > 0$. Hence, (2.3) holds. This, in combination with the fact that for every $x \in B$,

$$\begin{aligned}\|P_{A(x_B,r_B,j)}f\|_2 &\leq \mu((j + 1)B)^{1/2}\left(\fint_{(j+1)B} |f(y)|^2 d\mu(y)\right)^{1/2} \\ &\leq C(j + 1)^{n/2}\mu(B)^{1/2}\mathfrak{M}_2(f)(x), \qquad (2.5)\end{aligned}$$

yields that

$$\begin{aligned}\|P_B e^{-(2^{-k}-it)L}P_{X\setminus 2B}f\|_2 &\le C\sum_{j=2}^{\infty} j^{-(M-\frac{3n}{2})}\mu(B)^{1/2}\mathfrak{M}_2(f)(x)\\ &\le C\mu(B)^{1/2}\mathfrak{M}_2(f)(x)\end{aligned}$$

as long as we choose $M > 3n/2$ in (2.4). This proves Lemma 2.4. □

Lemma 2.5. *Let $m \ge 2$ and L satisfies the Davies-Gaffney estimates (DG_m). For a given $\phi \in C_0^\infty([\frac{1}{4}, 4])$, we write $\phi_e(\lambda) = e^\lambda\phi(\lambda)$. Then for every $M > 0, k \in \mathbb{N}^+$ and $s > 0$, there exists a constant $C = C(m, n, M)$ independent of k and s such that for every $j = 2, 3, \dots$*

$$\left\|P_B(I - e^{-sL})^K\phi_e(2^{-k}L)P_{A(x_B,r_B,j)}f\right\|_2 \le Cj^{-M}(2^{k/m}r_B)^{-M-n}\|P_{A(x_B,r_B,j)}f\|_2$$

for all $B \subset X$ with $r_B \ge c2^{(m-1)k/m}$ for some $c \ge 1/4$.

As a consequence, we have

$$\left\|P_B(I - e^{-sL})^K\phi_e(2^{-k}L)P_{X\setminus 2B}f\right\|_2 \le C\mu(B)^{1/2}\mathfrak{M}_2(f)(x).$$

Proof. We write

$$\|P_B(I - e^{-sL})^K\phi_e(2^{-k}L)P_{X\setminus 2B}f\|_2 \le \sum_{j=2}^{\infty}\|P_B(1 - e^{-sL})^K\phi_e(2^{-k}L)P_{A(x_B,r_B,j)}f\|_2.$$

Note that the function $(1 - e^{-s\lambda})^K e^{2^{-k}\lambda}\phi_k(\lambda)$ is supported in $[2^{k-2}, 2^{k+2}]$. We apply Lemma 2.2 with $R = 2^{k+2}$ to obtain that for every $M > 0$ and $j \ge 2$,

$$\begin{aligned}\left\|P_B(1 - e^{-sL})^K\phi_e(2^{-k}L)P_{A(x_B,r_B,j)}\right\|_{2\to 2} &\le Cj^{-M}(2^{k/m}r_B)^{-M-n}\|\delta_{2^{k+2}}((1 - e^{-s\lambda})^K e^{2^{-k}\lambda}\phi_k(\lambda))\|_{W_2^{M+n+1}}\\ &\le Cj^{-M}(2^{k/m}r_B)^{-M-n}\|(1 - e^{-2^{(k+2)}s\lambda})^K e^{4\lambda}\phi(4\lambda)\|_{W_2^{M+n+1}}\\ &\le Cj^{-M}(2^{k/m}r_B)^{-M-n}. \qquad (2.6)\end{aligned}$$

This, in combination with (2.5), yields that for every $x \in B$,

$$\begin{aligned}\|P_B(I - e^{-sL})^K\phi_e(2^{-k}L)P_{X\setminus 2B}f\|_2 &\le C\sum_{j=2}^{\infty} j^{-(M-\frac{n}{2})}(2^{k/m}r_B)^{-M-n}\mu(B)^{1/2}\mathfrak{M}_2(f)(x)\\ &\le C\mu(B)^{1/2}\mathfrak{M}_2(f)(x)\end{aligned}$$

as long as we choose $M > n/2$ in the first inequality above and notice the fact that $2^{k/m}r_B \ge 1/4$. This proves Lemma 2.5. □

Proof of Proposition 2.3. Let us show (2.2) when $d(B_1, B_2) \ge 6r_{B_1}$. By spectral theory, we write

$$(I - e^{-sL})^K e^{itL}\phi_k(L) = e^{-(2^{-k}-it)L}[(I - e^{-sL})^K\phi_e(2^{-k}L)] = S_{k,t}(L)T_k(L)$$

where we write $\phi_e(\lambda) = e^\lambda\phi(\lambda)$,

$$S_{k,t}(L) = e^{-(2^{-k}-it)L}$$

and

$$T_k(L) = (I - e^{-sL})^K\phi_e(2^{-k}L).$$

Set $G = \{x : \mathrm{dist}(x, B_1) \le d(B_1, B_2)/2\}$. Then it is clear that $\mathrm{dist}(B_2, \bar{G}) \ge d(B_1, B_2)/2$, where we use $\bar{G}$ to denote the topological closure of the set G. Moreover, from the definition of G, it is also clear that $\mathrm{dist}(X\backslash G, B_1) \ge d(B_1, B_2)/3$. Furthermore, based on the above observations we have

$$G \subset \bigcup_{j=\lfloor \frac{d(B_1,B_2)}{2r_{B_2}} \rfloor}^{\lfloor 2+\frac{d(B_1,B_2)}{r_{B_2}} \rfloor+1} A(x_{B_2}, r_{B_2}, j) \quad \text{and} \quad X\backslash G \subset \bigcup_{j=\lfloor \frac{d(B_1,B_2)}{2r_{B_1}} \rfloor-1}^{\infty} A(x_{B_1}, r_{B_1}, j),$$

where $\lfloor a \rfloor$ denotes the greatest integer that is smaller than a.

Then by noting that $S_{k,t}(L)$ is uniformly bounded on $L^2(X)$ and by Lemma 2.5,

$$\begin{aligned}
\left\| P_{B_1} S_{k,t}(L) \left(P_G T_k(L) P_{B_2} f \right) \right\|_2 &\le \left\| S_{k,t}(L) \left(P_G T_k(L) P_{B_2} f \right) \right\|_2 \\
&\le C \left\| P_G T_k(L) P_{B_2} f \right\|_2 \\
&\le C \sum_{j=\lfloor d(B_1,B_2)/(2r_{B_2}) \rfloor}^{\lfloor 2+d(B_1,B_2)/r_{B_2} \rfloor+1} \left\| P_{A(x_{B_2}, r_{B_2}, j)} T_k(L) P_{B_2} f \right\|_2 \\
&\le C \sum_{j=\lfloor d(B_1,B_2)/(2r_{B_2}) \rfloor}^{\lfloor 2+d(B_1,B_2)/r_{B_2} \rfloor+1} j^{-M} r_{B_2}^{-M} \left\| P_{B_2} f \right\|_2 \\
&\le C \left(1 + \frac{d(B_1, B_2)}{r_{B_2}} \right)^{-M+1} r_{B_2}^{-M} \left\| P_{B_2} f \right\|_2 \\
&\le C \left(1 + \frac{d(B_1, B_2)}{2^{(m-1)k/m}(1+|t|)} \right)^{-M+1} \left\| P_{B_2} f \right\|_2 \qquad (2.7)
\end{aligned}$$

for any $M > 0$, where in the last inequality we use the facts that $r_{B_2} \ge 1/4$ and that $d(B_1, B_2) > 2^{k(m-1)}(1 + |t|)$.

On the other hand, we apply Lemma 2.4 and the fact that $T_k(L)$ is uniformly bounded on $L^2(X)$ to see that for every $M > 0$,

$$\begin{aligned}
\left\| P_{B_1} S_{k,t}(L) \left(P_{X\backslash G} T_k(L) P_{B_2} f \right) \right\|_2 &\le \sum_{j=\lfloor d(B_1,B_2)/(2r_{B_1}) \rfloor-1}^{\infty} \left\| P_{B_1} S_{k,t}(L) P_{A(x_{B_1}, r_{B_1}, j)} \left(T_k(L) P_{B_2} f \right) \right\|_2 \\
&\le \sum_{j=\lfloor d(B_1,B_2)/(2r_{B_1}) \rfloor-1}^{\infty} j^{-M} \left(1 + \frac{r_{B_1}}{2^{(m-1)k/m}(1+|t|)} \right)^{-M} \left\| T_k(L) P_{B_2} f \right\|_2 \\
&\le C \left(1 + \frac{d(B_1, B_2)}{2^{(m-1)k/m}(1+|t|)} \right)^{-M} \left\| P_{B_2} f \right\|_2 \qquad (2.8)
\end{aligned}$$

Therefore, we combine the estimates (2.7) and (2.8) to obtain that for every $M > 0$,

$$\begin{aligned}
\left\| P_{B_1} S_{k,t}(L) \left(T_k(L) P_{B_2} \right) \right\|_2 &\le \left\| P_{B_1} S_{k,t}(L) \left(P_G T_{k,t}(L) P_{B_2} f \right) \right\|_2 \\
&\quad + \left\| P_{B_1} S_{k,t}(L) \left(P_{X\backslash G} T_k(L) P_{B_2} f \right) \right\|_2 \\
&\le C \left(1 + \frac{d(B_1, B_2)}{2^{(m-1)k/m}(1+|t|)} \right)^{-M} \left\| P_{B_2} f \right\|_2,
\end{aligned}$$

which shows that (2.2) holds. The proof of Proposition 2.3 is complete. □

In order to prove Theorem 1.3, we also need the following estimate for the operator $e^{itL}\phi_k(tL)$, $t>0$. Recall that $\phi\in C_0^\infty([1/4,4])$ is a cut-off function and $\phi_k(s)=\phi(2^{-k}s)$ for every $k\geq 1$.

Proposition 2.6. *Let $m\geq 2$ and L satisfies the Davies-Gaffney estimates (DG_m). For every $M>0$, $K\in\mathbb{N}^+$, $s>0, t>0$ and $k\geq 1$, there exists a constant $C=C(M,n,K)$ independent of t, s, and k such that*

$$\left\|P_{B_1}(I-e^{-sL})^K e^{itL}\phi_k(tL)P_{B_2}f\right\|_2\leq C\left(1+\frac{d(B_1,B_2)}{2^{(m-1)k/m}t^{1/m}}\right)^{-M}\|P_{B_2}f\|_2$$

for all $B_i\subset X$ with $r_{B_1}=r_{B_2}\geq c2^{(m-1)k/m}t^{1/m}$ for some $c\geq 1/4$.

Proof. The proof of Proposition 2.6 can be obtained by making minor modifications with the proof of Proposition 2.3, we leave the detail to the reader. □

2.2. Spectral multipliers. The following result is a standard known result in the theory of spectral multipliers of non-negative selfadjoint operators.

Proposition 2.7. *Let $m\geq 2$. Suppose that (X,d,μ) is a space of homogeneous type with a dimension n. Suppose that L satisfies the property ($\mathrm{GGE}_{\mathrm{p_0},\mathrm{p_0'},\mathrm{m}}$) for some $1\leq p_0<2$. Then we have*

(a) *Assume in addition that F is an even bounded Borel function such that $\sup_{R>0}\|\eta\delta_R F\|_{C^\beta}<\infty$ for some integer $\beta>n/2+1$ and some non-trivial function $\eta\in C_0^\infty(0,\infty)$. Then the operator $F(L)$ is bounded on $L^p(X)$ for all $p_0<p<p_0'$,*

$$\|F(L)\|_{p\to p}\leq C_\beta\left(\sup_{R>0}\|\eta\delta_R F\|_{C^\beta}+F(0)\right). \tag{2.9}$$

(b) *Fix a non-zero C^∞ bump function φ on $\mathbb{R}$ such that $\operatorname{supp}\phi\subseteq(1/2,2)$ for all $\lambda>0$ and set $\varphi_0(\lambda)=\sum_{\ell\leq 0}\varphi(2^{-\ell}\lambda)$ and $\varphi_k(\lambda)=\varphi(2^{-k}\lambda)$ for $k=1,2,\cdots$. Then for all $p_0<p<p_0'$,*

$$\left\|\Big(\sum_{k=0}^\infty|\varphi_k(L)f|^2\Big)^{1/2}\right\|_p\leq C_p\|f\|_p. \tag{2.10}$$

In addition, if $\sum_{k\geq 0}\varphi_k(\lambda)=1$ for all $\lambda>0$, then we have

$$\|f\|_p\cong C_p\left\|\Big(\sum_{k=0}^\infty|\varphi_k(L)f|^2\Big)^{1/2}\right\|_p,\quad p_0<p<p_0'. \tag{2.11}$$

Proof. Assertion (a) follows from [5, Theorem 1.1], see also [10, Lemma 4.5]. The proof of assertion (b) follows from Stein's classical proof [40, Chapter IV]. We give a brief argument of this proof for completeness and convenience for the reader.

Let us introduce the Rademacher function, which is defined as follows: i) The function $r_0(t)$ is defined by $r_0(t)=1$ on $[0,1/2]$ and $r_0(t)=-1$ on $(1/2,1)$, and then extended to $\mathbb{R}$ by periodicity; ii) For $k\in\mathbb{N}\backslash\{0\}$, $r_k(t)=r_0(2^kt)$. Define

$$F(t,\lambda)=\sum_{k=0}^\infty r_k(t)\varphi_k(\lambda).$$

A straightforward computation shows that for every integer $\beta > n/2+1$, $\sup_{R>0}\|\eta F(t,R\lambda)\|_{C^\beta} \le C_\beta$ uniformly in $t\in[0,1]$. Then we apply (2.9) to see that for all $p\in(p_0,p_0')$,

$$\|F(t,L)f\|_p = \Big\|\sum_{k=0}^{\infty} r_k(t)\varphi_k(L)f\Big\|_p \le C\|f\|_p$$

with $C>0$ uniformly in $t\in[0,1]$. This, in combination with the standard inequality for Rademacher functions:

$$\Big(\sum_{k=0}^{\infty}|\varphi_k(L)f|^2\Big)^{p/2} \cong \int_0^1 \Big|\sum_{k=0}^{\infty} r_k(t)\varphi_k(L)f\Big|^p dt,$$

yields

$$\Big\|\Big(\sum_{k=0}^{\infty}\big|\varphi_k(L)f\big|^2\Big)^{1/2}\Big\|_p \cong \left(\int_0^1 \Big\|\sum_{k=0}^{\infty} r_k(t)\varphi_k(L)f\Big\|_p^p dt\right)^{1/p} \le C_p\|f\|_p.$$

This proves (2.10).

When $\sum_{k\ge0}\varphi_k(\lambda)=1$ for all $\lambda>0$, it follows by the spectral theory [31] that $\sum_{k\ge0}\varphi_k(L)f=f$ for every $f\in L^2$. From it, we obtain (2.11) by using (2.10) and the standard duality argument (see for example, [40, Chapter IV]). This completes the proof of Proposition 2.7. □

3. Sharp endpoint L^p-Sobolev estimates for Scrödinger groups

In this section we prove (1.7) in Theorem 1.1. First, we note that from (1.12), estimate (1.7) holds for $s>n|1/2-1/p|$. By duality, it suffices to verify (1.7) for $2\le p<p_0'$ and $s=n|1/2-1/p|$. Also, it follows by the spectral theory [31] that (1.7) holds for $p=2$. For $p\neq2$, we recall that when L satisfies the generalized Gaussian estimates ($\mathrm{GGE}_{p_0,p_0',\mathrm{m}}$) for some $1\le p_0<2$, it was proved by Blunck [4, Theorem 1.1] that for every $z\in\mathbb{C}^+$,

$$\|e^{-zL}\|_{p\to p} \le C\left(\frac{|z|}{\mathrm{Re}\, z}\right)^{n|\frac12-\frac1p|} \tag{3.1}$$

for all $p\in[p_0,p_0']$ with $p\neq\infty$. From this, we have the following sharp L^p frequency truncated estimates for the Schrödinger group.

Proposition 3.1. *Suppose that (X,d,μ) is a space of homogeneous type with a dimension n. Suppose that L satisfies the property ($\mathrm{GGE}_{p_0,p_0',\mathrm{m}}$) for some $1\le p_0<2$. Then for every $p\in(p_0,p_0')$ and $k\ge0$,*

$$\|e^{itL}\phi(2^{-k}L)f\|_p \le C(1+2^k|t|)^s\|f\|_p,\quad t\in\mathbb{R},\quad s=n\Big|\frac12-\frac1p\Big| \tag{3.2}$$

uniformly for $t\in\mathbb{R}$ and for ϕ in bounded subsets of $C_0^\infty(\mathbb{R})$.

Proof. To show (3.2), we apply (3.1) with $z=2^{-k}-it$ to get that for every $\phi\in C_0^\infty(\mathbb{R})$,

$$\begin{aligned}\|e^{itL}\phi(2^{-k}L)\|_{p\to p} &= \Big\|e^{-(2^{-k}-it)L}[\phi_e(2^{-k}L)]\Big\|_{p\to p} \le C(1+2^k|t|)^s\|\phi_e(2^{-k}L)\|_{p\to p}\\ &\le C(1+2^k|t|)^s,\end{aligned}$$

where $\phi_e(\lambda) = e^{\lambda}\phi(\lambda)$. In the last inequality we used Proposition 2.7 to know that the operator $\phi_e(2^{-k}L)$ is bounded on $L^p(X)$ all $p \in (p_0, p_0')$. This completes the proof of Proposition 3.1. □

To prove Theorem 1.1, let us introduce some tools needed in the proof. Let T be a sublinear operator which is bounded on $L^2(X)$ and $\{A_r\}_{r>0}$ be a family of linear operators acting on $L^2(X)$. For $f \in L^2(X)$, we follow [2] to define

$$\mathfrak{M}^{\#}_{T,A} f(x) = \sup_{B \ni x} \left(\fint_B \left| T(I - A_{r_B}) f \right|^2 d\mu \right)^{1/2},$$

where the supremum is taken over all balls B in X containing x, and r_B is the radius of B. Then we have the following result. For its proof, we refer readers to [2, Lemma 2.3], [18, Lemma 5.4] and [37, Proposition 3.2].

Proposition 3.2. *Suppose that T is a sublinear operator which is bounded on $L^2(X)$ and that $q \in (2, \infty]$. Assume that $\{A_r\}_{r>0}$ is a family of linear operators acting on $L^2(X)$ and that*

$$\left(\fint_B |T A_{r_B} f(y)|^q d\mu(y) \right)^{1/q} \le C \mathfrak{M}_2(Tf)(x) \tag{3.3}$$

for all $f \in L^2(X)$, all $x \in X$ and all balls $B \ni x$, r_B being the radius of B.

Then for $0 < p < q$, there exists C_p such that

$$\left\| \mathfrak{M}_2(Tf) \right\|_p \le C_p \left(\left\| \mathfrak{M}^{\#}_{T,A} f \right\|_p + \|f\|_p \right) \tag{3.4}$$

for every $f \in L^2(X)$ for which the left-hand side is finite (if $\mu(X) = \infty$, the term $C_p \|f\|_p$ can be omitted in the right-hand side of (3.4)).

Proof of Theorem 1.1. Let us show Theorem 1.1 for $2 < p < p_0'$ and $s = n|1/2 - 1/p|$. We fix a non-zero C^∞ bump function φ on $\mathbb{R}$ such that

$$\operatorname{supp} \varphi \subseteq (\frac{1}{2}, 2) \ \text{ and } \ \sum_{\ell \in \mathbb{Z}} \varphi(2^{-\ell}\lambda) = 1 \quad \text{for all } \lambda > 0 \tag{3.5}$$

and set $\varphi_0(\lambda) = \sum_{\ell \le 0} \varphi(\lambda/2^\ell)$ and $\varphi_\ell(\lambda) = \varphi(\lambda/2^\ell)$ for $\ell = 1, 2, \ldots$.

For this fixed bump function φ, we consider an operator T_φ, given by

$$T_\varphi f(x) = \left(\sum_{k \ge 0} |\varphi_k(L) e^{itL} f(x)|^2 \right)^{1/2} \tag{3.6}$$

for every $f \in L^2(X)$. Then from (2.11), it is direct to see that $\|e^{itL} f\|_p \le C \|T_\varphi f\|_p$ for $2 < p < p_0'$.

Next, we define a sharp maximal function $\mathfrak{M}^{\#}_{T_\varphi, L, K}$ of T_φ as follows: for every $K \in \mathbb{N}$ and every $f \in L^2(X)$,

$$\mathfrak{M}^{\#}_{T_\varphi, L, K} f(x) \ = \ \sup_{B \ni x} \left(\fint_B \left| T_\varphi (I - e^{-r_B^m L})^K f(y) \right|^2 d\mu(y) \right)^{1/2}, \tag{3.7}$$

where the supremum is taken over all balls B in X containing x, and r_B is the radius of B. In order to prove Theorem 1.1, it suffices to show the following two arguments:

($\mathfrak{a}_1$) the operator T_φ satisfies condition (3.3) for every $2 < p < q < p_0'$ and $A_{r_B} = I - (I - e^{-r_B^m L})^K$ for every $K \in \mathbb{N}$;

($\mathfrak{a}_2$) by choosing K large enough, for $s = n|1/2 - 1/p|$,

$$\left\| \mathfrak{M}^{\#}_{T_\varphi, L, K} f \right\|_p \leq C(1+|t|)^s \left(\sum_{k \geq 0} 2^{ksp} \|\varphi_k(L) f\|_p^p \right)^{1/p}. \tag{3.8}$$

Before we prove the above two arguments ($\mathfrak{a}_1$) and ($\mathfrak{a}_2$), let us show that Theorem 1.1 is a straightforward consequence of them. Indeed, when ($\mathfrak{a}_1$) holds for T_φ, it follows from (b) of Proposition 2.7 and Proposition 3.2 that for $2 < p < p_0'$, $\|\mathfrak{M}_2(T_\varphi f)\|_p \leq C_p(\|f\|_p + \|\mathfrak{M}^{\#}_{T_\varphi, L, K} f\|_p)$. This, together with (3.8), yields that

$$\begin{aligned}
\|e^{itL} f\|_p \leq C\|T_\varphi f\|_p &\leq C\|\mathfrak{M}_2(T_\varphi f)\|_p \leq C_p(\|f\|_p + \|\mathfrak{M}^{\#}_{T_\varphi, L, K} f\|_p) \\
&\leq C\|f\|_p + C(1+|t|)^s \left(\sum_{k \geq 0} 2^{ksp} \|\varphi_k(L) f\|_p^p \right)^{1/p} \\
&\leq C\|f\|_p + C(1+|t|)^s \left(\|\varphi_0(f)\|_p + \left\| \left(\sum_{k>0} 2^{2ks} |\varphi_k(L) f|^2 \right)^{1/2} \right\|_p \right) \\
&\leq C(1+|t|)^s \left(\|f\|_p + \left\| \left(\sum_{k>0} \left| \phi_k(L)[L^s f] \right|^2 \right)^{1/2} \right\|_p \right) \\
&\leq C(1+|t|)^s \left(\|f\|_p + \|L^s f\|_p \right), \qquad (3.9)
\end{aligned}$$

where in the fifth inequality we have used the embedding $\ell^2 \hookrightarrow \ell^p$ for $p \geq 2$, in the sixth inequality the function $\phi_k(\lambda) = \varphi(2^{-k}\lambda)(2^{-k}\lambda)^{-s}$, and in the last inequality we used (b) of Proposition 2.7 for the Littlewood-Paley result for functions in $L^p(X)$. This proves Theorem 1.1.

We now first prove the argument ($\mathfrak{a}_1$). Indeed, in virtue of the formula

$$I - (I - e^{-r_B^m L})^K = \sum_{\tau=1}^{K} \binom{K}{\tau} (-1)^{\tau+1} e^{-\tau r_B^m L} \tag{3.10}$$

and the commutativity property $\varphi_k(L) e^{itL} e^{-\tau r_B^m L} = e^{-\tau r_B^m L} \varphi_k(L) e^{itL}$, it is enough to show that for all ball B containing x,

$$\left(\fint_B \left(\sum_{k \geq 0} \left| e^{-\tau r_B^m L} \varphi_k(L) e^{itL} f(y) \right|^2 \right)^{q/2} d\mu(y) \right)^{1/q} \leq C \mathfrak{M}_2(T_\varphi f)(x). \tag{3.11}$$

Let us prove (3.11). From hypothesis ($\mathrm{GGE}_{p_0, p_0', m}$), it is seen that condition ($\mathrm{GGE}_{2,q,m}$) holds for $2 < p < q < p_0'$, i.e, there exist constants $C, c > 0$ such that for every $u > 0$ and $x, y \in X$,

$$\left\| P_{B(x,u^{1/m})} e^{-uL} P_{B(y,u^{1/m})} \right\|_{2 \to q} \leq C V(x, u^{1/m})^{-(\frac{1}{2} - \frac{1}{q})} \exp\left(-c \left(\frac{d(x,y)^m}{u} \right)^{\frac{1}{m-1}} \right). \tag{3.12}$$

By Minkowski's inequality, (3.12) and (ii) of Lemma 2.1, conditions (1.5) and (2.1) for every $\tau = 1, 2, \dots, K$ and every ball B containing x, the left hand side of (3.11) is less than

$$\begin{aligned}
&V(B)^{-1/q} \sum_{j=0}^{\infty} \left(\sum_{k\geq 0} \left(\|P_B e^{-\tau r_B^m L} P_{A(x_B,r_B,j)} \varphi_k(L) e^{itL} f\|_q \right)^2 \right)^{1/2} \\
&\leq V(B)^{-1/q} \sum_{j=0}^{\infty} \|P_B e^{-\tau r_B^m L} P_{A(x_B,r_B,j)}\|_{2\to q} \left(\sum_{k\geq 0} \|\varphi_k(L) e^{itL} f\|^2_{L^2(A(x_B,r_B,j))} \right)^{1/2} \\
&\leq C \sum_{j=0}^{\infty} \left(\frac{V((j+1)B)}{V(B)} \right)^{1/2} e^{-c_\tau j^{m/(m-1)}} (1+j)^n \left(\fint_{(j+1)B} \sum_{k\geq 0} \left| \varphi_k(L) e^{itL} f(y) \right|^2 d\mu(y) \right)^{1/2} \\
&\leq C \sum_{j=0}^{\infty} e^{-c_\tau j^{m/(m-1)}} (1+j)^{3n/2} \mathfrak{M}_2(T_\varphi f)(x) \\
&\leq C \mathfrak{M}_2(T_\varphi f)(x).
\end{aligned}$$

The above estimate yields (3.11).

Thus, we obtain that the argument $(\mathfrak{a}_1)$ holds.

We now show the argument $(\mathfrak{a}_2)$. In the sequel we let $\phi \in C_0^\infty(\mathbb{R})$ supported in $(1/4, 4)$ and $\phi(x) = 1$ if $x \in (1/2, 2)$, and set $\phi_k(x) = \phi(2^{-k}x)$ for $k \geq 1$. Let $\phi_0 \in C_0^\infty([-4, 4])$ and $\phi_0(x) = 1$ if $x \in (-2, 2)$. By spectral theory, we have that $\varphi_k(L) f = \phi_k(L)\varphi_k(L) f$ for $k \geq 0$ and for every $f \in L^2(X)$. Hence, the proof of (3.8) reduces to show that

$$\|I\|_p + \|II\|_p + \|III\|_p \leq C(1+|t|)^s \left(\sum_{k\geq 0} \|\varphi_k(L) f\|_p^p \right)^{1/p}, \tag{3.13}$$

where

$$\begin{aligned}
I(x) &= \sup_{B\ni x} \left(\fint_B \sum_{0\leq k\leq -j} 2^{-2ks} \left| (I - e^{-r_B^m L})^K \phi_k(L) [e^{itL} \phi_k(L) \varphi_k(L) f](y) \right|^2 d\mu(y) \right)^{1/2}, \\
II(x) &= \sup_{B\ni x} \left(\fint_B \sum_{\substack{k+j>0 \\ j\geq (m-1)k + m\log_2(2+2|t|) \\ k\geq 0}} 2^{-2ks} \left| (I - e^{-r_B^m L})^K e^{itL} \phi_k(L) [\varphi_k(L) f](y) \right|^2 d\mu(y) \right)^{1/2}, \\
III(x) &= \sup_{B\ni x} \left(\fint_B \sum_{\substack{k+j>0 \\ j< (m-1)k + m\log_2(2+2|t|) \\ k\geq 0}} 2^{-2ks} \left| (I - e^{-r_B^m L})^K e^{itL} \phi_k(L) [\varphi_k(L) f](y) \right|^2 d\mu(y) \right)^{1/2}.
\end{aligned}$$

Here, we use the notation in the above decomposition that the ball B is centered at x_B and its radius r_B is in $[2^{(j-1)/m}, 2^{j/m})$ for some $j \in \mathbb{Z}$.

Estimate of the term $I(x)$. By the Minkowski inequality, we see that

$$\begin{aligned} I(x) &\le \sup_{B\ni x}\Big(\fint_B \big|(I-e^{-r_B^m L})^K e^{itL}\phi_0(L)[\varphi_0(L)f](y)\big|^2 d\mu(y)\Big)^{\frac12} \\ &\quad + \sup_{B\ni x}\mu(B)^{-1/2}\sum_{u=0}^{\infty}\sum_{1\le k\le -j} 2^{-ks}\big\|P_B(I-e^{-r_B^m L})^K\phi_k(L)P_{A(x_B,r_B,u)}[e^{itL}\phi_k(L)\varphi_k(L)f]\big\|_2 \\ &= I_1(x)+I_2(x). \end{aligned}$$

For the term $I_1(x)$, from the arguments in (3.10) and (3.11), it is direct to see that for every $x\in B$, $I_1(x)\le C\mathfrak{M}_2(e^{itL}\phi_0(L)[\varphi_0(L)f])(x)$. Then from Proposition 3.1,

$$\|I_1\|_p \le C\big\|e^{itL}\phi_0(L)[\varphi_0(L)f]\big\|_p \le C(1+|t|)^s\|\varphi_0(L)(f)\|_p.$$

For the term $I_2(x)$, since the function $(1-e^{-r_B^m\lambda})^K\phi_k(\lambda)$ is supported in $[2^{k-2},\ 2^{k+2}]$, $k\ge 1$, it tells us that for $u=0,1$,

$$\begin{aligned} \big\|P_B(I-e^{-r_B^m L})^K\phi_k(L)P_{A(x_B,r_B,u)}\big\|_{2\to 2} &\le \big\|(I-e^{-r_B^m L})^K\phi_k(L)\big\|_{2\to 2} \\ &\le C\|(1-e^{-r_B^m\lambda})^K\phi_k(\lambda)\|_{L^\infty} \\ &\le C\min\{1,(2^k r_B^m)^K\}, \end{aligned}$$

also for $u\ge 2$, we use Lemma 2.2 to obtain that for every $M>0$,

$$\begin{aligned} \big\|P_B(I-e^{-r_B^m L})^K\phi_k(L)P_{A(x_B,r_B,u)}\big\|_{2\to 2} &\le Cu^{-M}(2^{k/m}r_B)^{-M-n}\|\delta_{2^{k+2}}((1-e^{-r_B^m\lambda})^K\phi_k(\lambda))\|_{W_2^{M+n+1}} \\ &\le Cu^{-M}2^{-(k+j)(M+n)/m}\|(1-e^{-2^{(k+2)}r_B^m\lambda})^K\phi(4\lambda)\|_{W_2^{M+n+1}} \\ &\le Cu^{-M}\min\{2^{-(k+j)(M+n)/m}, 2^{(k+j)(K-M/m-n/m)}\}. \end{aligned} \tag{3.14}$$

Those, in combination with $k+j\le 0$ and the fact that for all $u\ge 0$ and $g\in L^2_{loc}(X)$

$$\begin{aligned} \|P_{A(x_B,r_B,u)}g\|_2 &\le \mu((u+1)B)^{1/2}\left(\fint_{(u+1)B}|g(y)|^2 d\mu(y)\right)^{1/2} \\ &\le C(1+u)^{n/2}\mu(B)^{1/2}\mathfrak{M}_2(g)(x), \end{aligned} \tag{3.15}$$

yield

$$\begin{aligned} I_2(x) &\le \sup_{B\ni x}\sum_{1\le k\le -j}\sum_{u=0}^{\infty} 2^{-ks}(1+u)^{-(M-n/2)}2^{(k+j)(K-(M+n)/m)}\mathfrak{M}_2\Big(e^{itL}\phi_k(L)\varphi_k(L)f\Big)(x) \\ &\le C\sup_{B\ni x}\sum_{1\le k\le -j} 2^{-ks}2^{(k+j)(K-(M+n)/m)}\mathfrak{M}_2\Big(e^{itL}\phi_k(L)\varphi_k(L)f\Big)(x), \end{aligned}$$

where $M>n/2$ and K is large enough so that $K>(M+n)/m$. We then use the embedding $\ell^p\hookrightarrow\ell^\infty$, the Minkowski inequality, $L^{p/2}$-boundedness of $\mathfrak{M}$ and Proposition 3.1 to see that

$$\begin{aligned} \|I_2\|_p &\le C\left\|\left(\sum_{j=-\infty}^{\infty}\left(\sum_{1\le k\le -j}2^{(k+j)(K-(M+n)/m)}2^{-ks}\mathfrak{M}_2\left(e^{itL}\phi_k(L)\varphi_k(L)f\right)(x)\right)^p\right)^{1/p}\right\|_p \\ &\le C\sum_{\ell\ge 0}2^{-\ell(K-(M+n)/m)}\left(\sum_{j<-\ell}2^{(\ell+j)sp}\left\|\mathfrak{M}_2\left(e^{itL}\phi_{-(\ell+j)}(L)\varphi_{-(\ell+j)}(L)f\right)\right\|_p^p\right)^{1/p} \end{aligned}$$

$$\begin{aligned}
&\le C\sum_{\ell\ge 0}2^{-\ell(K-(M+n)/m)}\left(\sum_{j<-\ell}2^{(\ell+j)sp}\left\|e^{itL}\phi_{-(\ell+j)}(L)[\varphi_{-(\ell+j)}(L)f]\right\|_p^p\right)^{1/p}\\
&\le C(1+|t|)^s\sum_{\ell\ge 0}2^{-\ell(K-(M+n)/m)}\left(\sum_{j<-\ell}\left\|\varphi_{-(\ell+j)}(L)f\right\|_p^p\right)^{1/p}\\
&\le C(1+|t|)^s\left(\sum_{k\ge 1}\|\varphi_k(L)f\|_p^p\right)^{1/p}
\end{aligned}$$

as desired, as long as K is chosen large enough so that $K > (M+n)/m$. Combining the estimates of I_1 and I_2 we get that

$$\|I\|_p \le C(1+|t|)^s\left(\sum_{k\ge 0}\|\varphi_k(L)f\|_p^p\right)^{1/p}.$$

Estimate of the term $II(x)$. Note that

$$\begin{aligned}
II(x) &\le \sup_{B\ni x}\left(\fint_B\left|(I-e^{-r_B^mL})^Ke^{itL}\phi_0(L)[\varphi_0(L)f](y)\right|^2d\mu(y)\right)^{\frac12}\\
&\quad+\sup_{B\ni x}\sum_{\substack{k+j>0\\ j\ge(m-1)k+m\log_2(2+2|t|)\\ k\ge1}}\sum_{\ell=0}^{\infty}2^{-ks}\mu(B)^{-1/2}\\
&\quad\times\left\|P_B(I-e^{-r_B^mL})^Ke^{itL}\phi_k(L)P_{A(x_B,r_B,\ell)}\right\|_{2\to2}\left\|P_{A(x_B,r_B,\ell)}[\varphi_k(L)f]\right\|_2\\
&= II_1(x)+II_2(x).
\end{aligned}$$

Similar to the estimate of $I_1(x)$ above, we see that $\|II_1\|_p \le C(1+|t|)^s\|\varphi_0(L)(f)\|_p$.

We now estimate $II_2(x)$. For a fixed $r_B > 0$, we choose a sequence of points $\{x_i\}_i \subset X$ such that $d(x_i,x_k) > r_B$ for $i \ne k$ and $\sup_{x\in X}\inf_i d(x,x_i) \le r_B$. Such sequence exists because X is separable. Set

$$J_\ell = \{B(x_i,r_B) : B(x_i,r_B)\cap A(x_B,r_B,\ell)\ne\emptyset\},\quad \ell\ge 0.$$

It follows from (1.6) that for every $B(x_i,r_B)\in J_\ell$,

$$V(x_B,r_B)\le\left(1+\frac{d(x_i,x_B)}{r_B}\right)^DV(x_i,r_B)\le C(1+\ell)^DV(x_i,r_B)$$

and so

$$\#J_\ell \le C(1+\ell)^D\times\frac{V(x_B,(\ell+1)r_B)}{V(x_B,r_B)}\le C(1+\ell)^{D+n}<\infty. \tag{3.16}$$

Then we have

$$\begin{aligned}
II_2(x) &\le \sup_{B\ni x}\sum_{\substack{k+j>0\\ j\ge(m-1)k+m\log_2(2+2|t|)\\ k\ge1}}\sum_{\ell=0}^{\infty}\sum_{B(x_i,r_B)\in J_\ell}2^{-ks}\mu(B)^{-1/2}\\
&\quad\times\left\|P_B(I-e^{-r_B^mL})^Ke^{itL}\phi_k(L)P_{B(x_i,r_B)}\right\|_{2\to2}\left\|P_{A(x_B,r_B,\ell)}[\varphi_k(L)f]\right\|_2.
\end{aligned}$$

In this case, since $j \geq (m-1)k+m\log_2(2+2|t|)$ and so $r_B \geq c2^{(m-1)k/m}(1+|t|)$ with $c = 2^{(m-1)/m} \geq 1/4$, we apply Proposition 2.3 to see that for every $B(x_i,\, r_B) \in J_\ell$ with $\ell \geq 7, 8, \cdots,$

$$\left\|P_B(I-e^{-r_B^m L})^K e^{itL}\phi_k(L)P_{B(x_i,r_B)}\right\|_{2\to 2} \leq C\left(1+\frac{d(B,\, B(x_i,\, r_B))}{2^{(m-1)k/m}(1+|t|)}\right)^{-M} \leq C(1+\ell)^{-M} \tag{3.17}$$

for every $M > 0$. For $\ell = 0, 1, \cdots, 6$, it follows from L^2-boundedness of $(I-e^{-r_B^m L})^K e^{itL}\phi_k(L)$ that $\left\|P_B(I-e^{-r_B^m L})^K e^{itL}\phi_k(L)P_{B(x_i,r_B)}\right\|_{2\to 2} \leq C$. These, in combination with the fact that for every $x \in B$,

$$\begin{aligned}\|P_{A(x_B,r_B,\ell)}[\varphi_k(L)f]\|_2 &\leq \mu((\ell+1)B)^{1/2}\left(\fint_{(\ell+1)B}|\varphi_k(L)f(y)|^2 d\mu(y)\right)^{1/2}\\ &\leq C(\ell+1)^{n/2}\mu(B)^{1/2}\mathfrak{M}_2\left(\varphi_k(L)f\right)(x),\end{aligned}$$

imply

$$\begin{aligned}II_2(x) &\leq C\sum_{k\geq 1}\sum_{\ell=0}^{\infty}2^{-ks}(1+\ell)^{-(M-D-3n/2)}\mathfrak{M}_2\left(\varphi_k(L)f\right)(x)\\ &\leq C\sum_{k\geq 1}2^{-ks}\mathfrak{M}_2\left(\varphi_k(L)f\right)(x)\end{aligned}$$

as long as M in (3.17) is chosen large enough so that $M > D + 2n$. As a consequence, we have that for $2 < p < p_0'$,

$$\|II_2\|_p \leq C\left\|\sum_{k\geq 1}2^{-ks}\mathfrak{M}_2\left(\varphi_k(L)f\right)\right\|_p \leq C\left(\sum_{k\geq 1}\left\|\mathfrak{M}_2\big(\varphi_k(L)f\big)\right\|_p^p\right)^{1/p} \leq C\left(\sum_{k\geq 1}\|\varphi_k(L)f\|_p^p\right)^{1/p}.$$

Combining the estimates of II_1 and II_2 we obtain the estimate of II as desired.

Estimate of the term $III(x)$. As to be seen later, the term $III(x)$ is the major one.

Similar to the estimates for II and I above, we write

$$\begin{aligned}III(x) \leq{}& \sup_{B\ni x}\left(\fint_B\left|(I-e^{-r_B^m L})^K e^{itL}\phi_0(L)[\varphi_0(L)f](y)\right|^2 d\mu(y)\right)^{\frac{1}{2}}\\ &+\sup_{B\ni x}\left(\fint_B\sum_{\substack{k+j>0\\ j<(m-1)k+m\log_2(2+2|t|)\\ k\geq 1}}2^{-2ks}\left|(I-e^{-r_B^m L})^K e^{itL}\phi_k(L)[\varphi_k(L)f](y)\right|^2 d\mu(y)\right)^{1/2}\\ ={}& III_1(x)+III_2(x).\end{aligned}$$

Again, it is clear that $\|III_1\|_p \leq C(1+|t|)^s\|\varphi_0(L)(f)\|_p$. It suffices to verify $III_2(x)$.

For a given $x \in X$ and a ball $x \in B_j = B(x_{B_j}, r_{B_j})$ with $r_{B_j}^m \in [2^{j-1}, 2^j]$. We define a family of operators $\{A_{r_{B_j}}\}_{j=1}^\infty$ with non-negative kernels $\{a_{r_{B_j}}(x,y)\}_{j=1}^\infty$ such that

$$a_{r_{B_j}}(x,y) = \frac{1}{\mu(B(x,2r_{B_j}))}\chi_{B(x,2r_{B_j})}(y).$$

We will use

$$A_{r_{B_j}}g(x) = \int_X a_{r_{B_j}}(x,y)g(y)d\mu(y)$$

to replace the mean value $\fint_{B_j}$ in the term $III_2(x)$. It is seen that for every non-negative function $g \in L^1_{\rm loc}(X)$ and B_j containing x,

$$\fint_{B_j} g(y)d\mu(y) \;\le\; \left(\frac{\mu(B(x_{B_j}, 3r_{B_j}))}{\mu(B_j)}\right) A_{r_{B_j}} g(x) \le C A_{r_{B_j}} g(x)$$

and so $III_2(x) \le C\widetilde{III}_2(x)$, where

$$\widetilde{III}_2(x) := \sup_{j\in\mathbb{Z}} \left(\sum_{\substack{k+j>0 \\ j<(m-1)k+m\log_2(2+2|t|) \\ k\ge 1}} 2^{-2ks} A_{r_{B_j}} \left(\left| (I - e^{-r_{B_j}^m L})^K e^{itL} \phi_k(L)[\varphi_k(L) f] \right|^2 \right)(x) \right)^{1/2}. \tag{3.18}$$

Now for every $k \ge 1$, we choose a sequence $(x_\tau^{(k)})_\tau \in X$ such that $d(x_\tau^{(k)}, x_\ell^{(k)}) > 2^{k(m-1)/m}(1+|t|)$ for $\tau \ne \ell$ and $\sup_{x\in X} \inf_\tau d(x, x_\tau^{(k)}) \le 2^{k(m-1)/m}(1+|t|)$. Such sequence exists because X is separable. Let $B_\tau^{(k),*} = B(x_\tau^{(k)}, 8\cdot 2^{k(m-1)/m}(1+|t|))$ and define $B_\tau^{(k)}$ by the formula

$$B_\tau^{(k)} = \bar{B}\left(x_\tau^{(k)}, 2^{k(m-1)/m}(1+|t|)\right) \setminus \bigcup_{\ell<\tau} \bar{B}\left(x_\ell^{(k)}, 2^{k(m-1)/m}(1+|t|)\right),$$

where $\bar{B}\left(x_\tau^{(k)}, r\right) = \{y \in X\colon d(x_\tau^{(k)}, y) \le r\}$. We cover X by a grid $\mathscr{R}_k$ consisting of such $\{B_\tau^{(k)}\}_\tau$, that is, $X = \bigcup_{B_\tau^{(k)}\in\mathscr{R}_k} B_\tau^{(k)}$. For every $B_\tau^{(k)} \in \mathscr{R}_k$, we denote by $f^{B_\tau^{(k)}} = f\chi_{B_\tau^{(k)}}$. Hence, one writes

$$\widetilde{III}_2(x) \le III_{21}(x) + III_{22}(x), \tag{3.19}$$

where

$$III_{21}(x) = \sup_{j\in\mathbb{Z}} \left(\sum_{\substack{k+j>0 \\ j<(m-1)k+m\log_2(2+2|t|) \\ k\ge 1}} 2^{-2ks} A_{r_{B_j}} \left(\left| \sum_{B_\tau^{(k)}\in\mathscr{R}_k} \chi_{B_\tau^{(k),*}} (I - e^{-r_{B_j}^m L})^K e^{itL} \phi_k(L)[\varphi_k(L) f]^{B_\tau^{(k)}} \right|^2 \right)(x) \right)^{1/2}$$

and $III_{22}(x)$ is the analogous expression where $\chi_{B_\tau^{(k),*}}$ is replaced with $\chi_{X\setminus B_\tau^{(k),*}}$.

Let us first estimate the term $III_{21}(x)$. Using the embedding $\ell^p \to \ell^\infty$, the bounded overlap of $B_\tau^{(k),*}$ and Minkowski's inequality, we obtain that the L^p-norm of the term $III_{21}(x)$ is less than

$$C \left(\sum_{j\in\mathbb{Z}} \left\| \left(\sum_{\substack{k+j>0 \\ j<(m-1)k+m\log_2(2+2|t|) \\ k\ge 1}} 2^{-2ks} A_{r_{B_j}} \left(\sum_{B_\tau^{(k)}\in\mathscr{R}_k} \chi_{B_\tau^{(k),*}} \left| (I - e^{-r_{B_j}^m L})^K e^{itL} \phi_k(L)[\varphi_k(L) f]^{B_\tau^{(k)}} \right|^2 \right) \right)^{1/2} \right\|_p^p \right)^{1/p}.$$

To continue, we claim that the supports of the functions $\{A_{r_{B_j}}(\chi_{B_\tau^{(k),*}})\}_\tau$ have bounded overlap, uniformly in k. Assume this at the moment. Then by setting $\ell = k + j > 0$, applying Minkowski's inequality, and the above claim, we obtain that

$$\|III_{21}\|_p \le \sum_{\ell>0} E_\ell,$$

where

$$E_\ell := \left(\sum_{j<\ell} \sum_{B_\tau^{(\ell-j)} \in \mathscr{R}_{\ell-j}} 2^{-(\ell-j)sp} \left\| A_{r_{B_j}} \chi_{B_\tau^{(\ell-j),*}} \left(\left| (I - e^{-r_{B_j}^m L})^K e^{itL} \phi_{\ell-j}(L) \left[\varphi_{\ell-j}(L) f \right]^{B_\tau^{(\ell-j)}} \right|^2 \right) \right\|_{p/2}^{p/2} \right)^{1/p}.$$

We now show the claim. Note that for $B_\tau^{(k)} \in \mathscr{R}_k$, $B_\tau^{(k),*}$ has radius $8 \cdot 2^{k(m-1)/m}(1+|t|)$. It follows from $r_{B_j} \le 2^{j/m} \le 2^{k(m-1)/m-2}(1+|t|)$ that for fixed k, $A_{r_{B_j}}(\chi_{B_\tau^{(k),*}})(x) \cdot A_{r_{B_j}}(\chi_{B_\ell^{(k),*}})(x) = 0$ when $d(x_\tau^{(k)}, x_\ell^{(k)}) \ge 20 \cdot 2^{k(m-1)/m}(1+|t|)$. From (1.6), we know that

$$\begin{aligned} V(x_\ell^{(k)}, 2^{k(m-1)/m}(1+|t|)) &\le \left(1 + \frac{d(x_\ell^{(k)}, x_\tau^{(k)})}{r_B} \right)^D V(x_\tau^{(k)}, 2^{k(m-1)/m}(1+|t|)) \\ &\le C V(x_\tau^{(k)}, 2^{k(m-1)/m}(1+|t|)), \end{aligned}$$

which implies

$$\sup_\tau \#\{\ell : \ d(x_\tau^{(k)}, x_\ell^{(k)}) \le 30 \cdot 2^{\frac{k(m-1)}{m}}(1+|t|)\} \le \sup_x \frac{V(x, 30 \cdot 2^{\frac{k(m-1)}{m}}(1+|t|))}{V(x, 2^{\frac{k(m-1)}{m}-2}(1+|t|))} \le C < \infty.$$

Next we will show that

$$E_\ell \le C(1+|t|)^s 2^{-\ell s/m} \left(\sum_{k>0} \|\varphi_k(L) f\|_p^p \right)^{1/p}. \tag{3.20}$$

Once (3.20) is proven, we see that

$$\|III_{21}\|_p \le C(1+|t|)^s \left(\sum_{k\ge 1} \|\varphi_k(L) f\|_p^p \right)^{1/p}. \tag{3.21}$$

Let us prove estimate (3.20). First, we observe that for every $g \in L^1(X)$ and $p/2 > 1$,

$$\begin{aligned} \|A_{r_{B_j}} \left(\chi_{B_\tau^{(\ell-j),*}} g \right)\|_{p/2} &\le \left(\sup_{y \in B_\tau^{(\ell-j),*}} \int_X a_{r_{B_j}}^{p/2}(x,y) \chi_{B_\tau^{(\ell-j),*}}(y) d\mu(x) \right)^{2/p} \|g\|_1 \\ &\le C \sup_{y \in B_\tau^{(\ell-j),*}} [V(y, r_{B_j})^{-(1-\frac{2}{p})}] \|g\|_1. \end{aligned} \tag{3.22}$$

From this, we see that the term E_ℓ is dominated by a constant multiple of

$$\left(\sum_{j<\ell} \sum_{B_\tau^{(\ell-j)} \in \mathscr{R}_{\ell-j}} 2^{-(\ell-j)sp} \sup_{y \in B_\tau^{(\ell-j),*}} [V(y, r_{B_j})^{-(p/2-1)}] \left\| (I - e^{-r_{B_j}^m L})^K e^{itL} \phi_{\ell-j}(L) [\varphi_{\ell-j}(L) f]^{B_\tau^{(\ell-j)}} \right\|_2^p \right)^{1/p}.$$

Since the operator $(I - e^{-r_{B_j}^m L})^K e^{itL} \phi_{\ell-j}(L)$ is uniformly bounded on $L^2(X)$ and $[\varphi_{\ell-j}(L) f]^{B_\tau^{(\ell-j)}}$ is supported on the ball $B_\tau^{(\ell-j)}$, we see by the Hölder inequality that the term E_ℓ is controlled by a constant multiple of

$$\left(\sum_{j<\ell} 2^{-(\ell-j)sp} \sum_{B_\tau^{(\ell-j)} \in \mathscr{R}_{\ell-j}} \sup_{y \in B_\tau^{(\ell-j),*}} \left(\frac{\mu(B_\tau^{(\ell-j)})}{\mu(B(y, r_{B_j}))} \right)^{\frac{p}{2}-1} \left\| \left[\varphi_{\ell-j}(L) f \right]^{B_\tau^{(\ell-j)}} \right\|_p^p \right)^{1/p}.$$

Note that for $y \in B_\tau^{(\ell-j),*}$,

$$\left(\frac{\mu(B_\tau^{(\ell-j)})}{\mu(B(y, r_{B_j}))}\right) \le C(1+|t|)^n 2^{\frac{n}{m}[(\ell-j)(m-1)-j]},$$

which yields

$$\begin{aligned} E_\ell &\le C(1+|t|)^{n(\frac{1}{2}-\frac{1}{p})}\left(\sum_{j<\ell} 2^{-n(\ell-j)(\frac{p}{2}-1)} 2^{\frac{n}{m}[(\ell-j)(m-1)-j](\frac{p}{2}-1)} \sum_{B_\tau^{(\ell-j)}\in\mathscr{R}_{\ell-j}} \left\| \left[\varphi_{\ell-j}(L)f\right]^{B_\tau^{(\ell-j)}} \right\|_p^p \right)^{1/p} \\ &= C(1+|t|)^s 2^{-\ell s/m}\left(\sum_{j<\ell} \sum_{B_\tau^{(\ell-j)}\in\mathscr{R}_{\ell-j}} \left\| \left[\varphi_{\ell-j}(L)f\right]^{B_\tau^{(\ell-j)}} \right\|_p^p \right)^{1/p}. \end{aligned}$$

After summation in $B_\tau^{(\ell-j)} \in \mathscr{R}_{\ell-j}$, we obtain

$$E_\ell \le C(1+|t|)^s 2^{-\ell s/m}\left(\sum_{j<\ell} \left\|\varphi_{\ell-j}(L)f\right\|_p^p\right)^{1/p} \le C(1+|t|)^s 2^{-\ell s/m}\left(\sum_{k\ge 1} \|\varphi_k(L)f\|_p^p\right)^{1/p}.$$

This finishes the proof of (3.20) and concludes the desired estimate (3.21) for the term III_{21}.

Concerning the term III_{22}, we use the embedding $\ell^p \to \ell^\infty$ and the Minkowski inequality to see that the term $\|III_{22}\|_p$ is controlled by

$$\left(\sum_{j\in\mathbb{Z}}\left[\sum_{\substack{k+j>0\\ j<(m-1)k+m\log_2(2+2|t|)\\ k\ge 1}} 2^{-2ks}\left\|A_{r_{B_j}}\left(\left|\sum_{B_\tau^{(k)}\in\mathscr{R}_k} \chi_{X\setminus B_\tau^{(\ell-j),*}}(I-e^{-r_{B_j}^m L})^K e^{itL}\phi_k(L)[\varphi_k(L)f]^{B_\tau^{(k)}}\right|^2\right)\right\|_{p/2}\right]^{p/2}\right)^{1/p}.$$

The proof of Theorem 1.1 will be done if we can show that

$$\begin{aligned} &\left\|A_{r_{B_j}}\left(\left|\sum_{B_\tau^{(k)}\in\mathscr{R}_k} \chi_{X\setminus B_\tau^{(k),*}}(I-e^{-r_{B_j}^m L})^K e^{itL}\phi_k(L)\left[\varphi_k(L)f\right]^{B_\tau^{(k)}}\right|^2\right)\right\|_{p/2} \\ &\qquad \le C(1+|t|)^{n(1-\frac{2}{p})} 2^{\frac{n}{m}[k(m-1)-j](1-\frac{2}{p})} \|\varphi_k(L)f\|_p^2 \end{aligned} \tag{3.23}$$

since from it, we recall that $s = n|1/2 - 1/p|$ to see that

$$\begin{aligned} \|III_{22}\|_p &\le C(1+|t|)^s\left(\sum_{j\in\mathbb{Z}}\left[\sum_{\substack{k+j>0\\ j<(m-1)k+m\log_2(2+2|t|)\\ k\ge 1}} 2^{-2ks} 2^{\frac{n}{m}[k(m-1)-j](1-\frac{2}{p})} \|\varphi_k(L)f\|_p^2\right]^{p/2}\right)^{1/p} \\ &\le C(1+|t|)^s \sum_{\ell>0}\left(\sum_{j<\ell} 2^{-n(\ell-j)(\frac{p}{2}-1)} 2^{\frac{n}{m}[(\ell-j)(m-1)-j](\frac{p}{2}-1)} \left\|\varphi_{\ell-j}(L)f\right\|_p^p\right)^{1/p} \\ &= C(1+|t|)^s \sum_{\ell>0} 2^{-\ell s/m}\left(\sum_{j<\ell} \left\|\varphi_{\ell-j}(L)f\right\|_p^p\right)^{1/p} \end{aligned}$$

$$\leq \quad C(1+|t|)^s \left(\sum_{k\geq 1} \|\varphi_k(L)f\|_p^p \right)^{1/p}. \tag{3.24}$$

It remains to prove (3.23). Observe that $j < (m-1)k + m\log_2(2+2|t|)$, and $r_{B_j} \leq 2^{(m-1)k/m+1}(1+|t|)$. Fix $x \in X$, $k \geq 1$ and $j \in \mathbb{Z}$, we consider the following three cases of $x_\tau^{(k)}$:

Case 1: $d(x_\tau^{(k)}, x) \leq 6 \cdot 2^{(m-1)k/m}(1+|t|)$.

In this case, for any $z \in B(x, 2r_{B_j})$,

$$d(z, x_\tau^{(k)}) \leq d(z,x) + d(x_\tau^{(k)}, x) \leq 8 \cdot 2^{(m-1)k/m}(1+|t|);$$

and so $B(x, 2r_{B_j}) \cap \left(X \backslash B_\tau^{(k),*} \right) = \emptyset$;

Case 2: $d(x_\tau^{(k)}, x) \geq 10 \cdot 2^{(m-1)k/m}(1+|t|)$.

In this case, for any $z \in B(x, 2r_{B_j})$

$$d(z, x_\tau^{(k)}) \geq d(x_\tau^{(k)}, x) - d(z,x) \geq 8 \cdot 2^{(m-1)k/m}(1+|t|),$$

and so $B(x, 2r_{B_j}) \subseteq X \backslash B_\tau^{(k),*}$;

Case 3: $6 \cdot 2^{(m-1)k/m}(1+|t|) \leq d(x_\tau^{(k)}, x) \leq 10 \cdot 2^{(m-1)k/m}(1+|t|)$.

In this case, we see that $d(B_\tau^{(k)}, B(x, 2r_{B_j})) \geq 2^{(m-1)k/m}(1+|t|)$, and

$$\begin{aligned} &\sharp \left\{ \tau : 6 \cdot 2^{(m-1)k/m}(1+|t|) \leq d(x_\tau^{(k)}, x) \leq 10 \cdot 2^{(m-1)k/m}(1+|t|) \right\} \\ &\leq \quad \sup_x \frac{V(x, 10 \cdot 2^{(m-1)k/m+1}(1+|t|))}{V(x, 2^{(m-1)k/m-2}(1+|t|))} \leq C < \infty. \end{aligned} \tag{3.25}$$

From *Cases 1, 2* and *3*, we see that there exists a constant $C > 0$ independent of x and j such that

$$A_{r_{B_j}} \left(\left| \sum_{B_\tau^{(k)} \in \mathscr{R}_k} \chi_{X \backslash B_\tau^{(k),*}} (I - e^{-r_{B_j}^m L})^K e^{itL} \phi_k(L) \left[\varphi_k(L) f \right]^{B_\tau^{(k)}} \right|^2 \right)(x) \leq D_1(x) + C D_2(x),$$

where

$$D_1(x) : \quad = \quad A_{r_{B_j}} \left(\left| (I - e^{-r_{B_j}^m L})^K e^{itL} \phi_k(L) \left(\sum_{\tau:\, d(x_\tau^{(k)}, x) \geq 10 \cdot 2^{(m-1)k/m}(1+|t|)} \left[\varphi_k(L) f \right]^{B_\tau^{(k)}} \right) \right|^2 \right)(x)$$

and

$$D_2(x) := \left(\sum_{\tau:\, 6 \cdot 2^{(m-1)k/m}(1+|t|) \leq d(x_\tau^k, x) \leq 10 \cdot 2^{(m-1)k/m}(1+|t|)} A_{r_{B_j}} \left(\left| (I - e^{-r_{B_j}^m L})^K e^{itL} \phi_k(L) \left[\varphi_k(L) f \right]^{B_\tau^{(k)}} \right|^2 \right) \right)(x).$$

Let us estimate the term $D_1(x)$ by adapting an argument as in the term E_ℓ. First note that

$$X = \bigcup_{B_{\tau_1}^{(k)} \in \mathscr{R}_k} B_{\tau_1}^{(k)}.$$

Then we write

$$D_1(x) \quad \leq \quad \sum_{B_{\tau_1}^{(k)} \in \mathscr{R}_k} A_{r_{B_j}} \left(P_{B_{\tau_1}^{(k)}} \left| (I - e^{-r_{B_j}^m L})^K e^{itL} \phi_k(L) \left(\sum_{\tau:\, d(x_\tau^{(k)}, x) \geq 10 \cdot 2^{(m-1)k/m}(1+|t|)} \left[\varphi_k(L) f \right]^{B_\tau^{(k)}} \right) \right|^2 \right)(x).$$

Applying (3.22), we see that the $L^{p/2}$-norm of $D_1(x)$ is dominated by a constant times

$$\sum_{B^{(k)}_{\tau_1}\in\mathscr{R}_k}\sup_{y\in B^{(k)}_{\tau_1}}[V(y,r_{B_j})]^{-(1-\frac{2}{p})}\left\|P_{B^{(k)}_{\tau_1}}(I-e^{-r^m_{B_j}L})^K e^{itL}\phi_k(L)\left(\sum_{\tau:\, d(x^{(k)}_\tau,x^{(k)}_{\tau_1})\geq 10\cdot 2^{(m-1)k/m}(1+|t|)}[\varphi_k(L)f]^{B^{(k)}_\tau}\right)\right\|_2^2.$$

Observe that for every $B^{(k)}_{\tau_1}\in\mathscr{R}_k$,

- If $y\in B^{(k)}_{\tau_1}$, then

$$\left(\frac{\mu(B^{(k)}_\tau)}{V(y,r_{B_j})}\right) = \left(\frac{\mu(B^{(k)}_\tau)}{\mu(B^{(k)}_{\tau_1})}\right)\times\left(\frac{\mu(B^{(k)}_{\tau_1})}{V(y,r_{B_j})}\right)\leq C(1+|t|)^n 2^{n[k(m-1)-j]/m}\left(1+\frac{d(B^{(k)}_{\tau_1},B^{(k)}_\tau)}{2^{(m-1)k/m}(1+|t|)}\right)^D.$$

- A simple calculation shows that

$$\sharp\left\{\tau: 2^{\frac{(m-1)k}{m}+u}(1+|t|)\leq d(B^{(k)}_{\tau_1},B^{(k)}_\tau)\leq 2^{\frac{(m-1)k}{m}+u+1}(1+|t|)\right\}\leq C2^{u(D+n)}$$

and so

$$\begin{aligned}&\sum_{\tau:d(B^{(k)}_{\tau_1},B^{(k)}_\tau)>10\cdot 2^{(m-1)k/m}(1+|t|)}\left(1+\frac{d(B^{(k)}_{\tau_1},B^{(k)}_\tau)}{2^{(m-1)k/m}(1+|t|)}\right)^{-M}\\ &\leq \sum_{u=2}^\infty\sum_{\tau:2^{\frac{(m-1)k}{m}+u}(1+|t|)\leq d(B^{(k)}_{\tau_1},B^{(k)}_\tau)\leq 2^{\frac{(m-1)k}{m}+u+1}(1+|t|)}2^{-uM}\leq C\sum_{u=2}^\infty 2^{-u(M-(D+n))}\leq C\end{aligned}\tag{3.26}$$

for $M>D+n$. Since the function $[\varphi_k(L)f]^{B^{(k)}_\tau}$ is supported on the ball $B^{(k)}_\tau$, we apply Proposition 2.3 with $M>D+n$ and the Hölder inequality to see that $\|D_1\|_{p/2}$ is controlled by a constant multiple of

$$(1+|t|)^{n(1-\frac{2}{p})}2^{\frac{n}{m}[k(m-1)-j](1-\frac{2}{p})}\sum_{B^{(k)}_{\tau_1}\in\mathscr{R}_k}\sum_{\tau:\, d(x^{(k)}_\tau,x^{(k)}_{\tau_1})\geq 10\cdot 2^{(m-1)k/m}(1+|t|)}\left(1+\frac{d(B^{(k)}_{\tau_1},B^{(k)}_\tau)}{2^{(m-1)k/m}(1+|t|)}\right)^{-M}\left\|[\varphi_k(L)f]^{B^{(k)}_\tau}\right\|_p^p.$$

Changing the order of the summation for τ_1 and τ and by (3.26), we obtain

$$\|D_1\|_{p/2} \leq C(1+|t|)^{n(1-\frac{2}{p})}2^{\frac{n}{m}[k(m-1)-j](1-\frac{2}{p})}\|\varphi_k(L)f\|_p^p.$$

For the term D_2, we follow the similar approach as above in $D_1(x)$ to show that for every τ with $6\cdot 2^{(m-1)k/m}(1+|t|)\leq d(x^{(k)}_\tau,x)\leq 10\cdot 2^{(m-1)k/m}(1+|t|)$,

$$\left\|A_{r_{B_j}}\left(\left|(I-e^{-r^m_{B_j}L})^K e^{itL}\phi_k(L)\,[\varphi_k(L)f]^{B^{(k)}_\tau}\right|^2\right)\right\|_{p/2}\leq (1+|t|)^{n(1-\frac{2}{p})}2^{\frac{n}{m}[k(m-1)-j](1-\frac{2}{p})}\|\varphi_k(L)f\|_p^p,$$

and so by (3.25) in *Case 3*, we have that $\|D_2\|_{p/2}\leq (1+|t|)^{n(1-\frac{2}{p})}2^{\frac{n}{m}[k(m-1)-j](1-\frac{2}{p})}\|\varphi_k(L)f\|_p^p$. This finishes the proof of (3.23) and thereby (3.24) for the term III_{22} and concludes that

$$\|III_2\|_p\leq C(1+|t|)^{n(\frac{1}{2}-\frac{1}{p})}\left(\sum_{k\geq 1}\|\varphi_k(L)f\|_p^p\right)^{1/p}.$$

Combining the estimates of $III_1(x)$ and $III_2(x)$, we obtain the estimate for $III(x)$ as desired.

Finally, we combine estimates of I, II and III to obtain the estimate (3.8), and complete the proof of Theorem 1.1.

Proof of Corollary 1.2. The proof of Corollary 1.2 can be obtained by making a minor modifications with [34, Theorem 7.12], and we skip it here. □

We mention that our Theorem 1.1 can also apply to prove existence of solution (in L^p spaces) to the Schrödinger equation with initial data f in the domain of some power of the operator L. It can also be formulated in terms of generation of C-regularized groups. We will not develop this here, we refer the reader to de Laubenfels [15] and Ouhabaz's monograph [34, Chapter 7].

4. An application to Riesz means of the solutions of the Schrödinger equations

The aim of this section is to prove Theorem 1.3. Recall that when L is the Laplacian on the Euclidean spaces $\mathbb{R}^n$, the Riesz mean $I_s(t)(\Delta)$ in (1.9) was studied by Sjöstrand [39] . It was shown that $I_s(t)(\Delta)$ is uniformly bounded in $t \in \mathbb{R}\backslash\{0\}$ for $s > n|1/2 - 1/p|$, and they are unbounded for $s < n|1/2 - 1/p|$. The result was generalized to Lie groups and Riemannian manifolds by Lohoue[29] and by Alexopoulos [1]. In the abstract setting of operators on metric measure space, this result was extended by Carron, Coulhon and Ouhabaz [9] for operators with the Gaussian upper bounds, and by Blunck [4] for generalized Gaussian estimates for the operators. More precisely, the work of Blunck [4, Proposition A] shows that under the assumption of generalized Gaussian estimate ($\mathrm{GGE}_{p_0,p_0',m}$) for some $1 \le p_0 < 2$, then the Riesz means operator $I_s(t)(L)$ is bounded on $L^p(X)$ uniformly for all $t \in \mathbb{R}\backslash\{0\}$, $p \in (p_0, p_0')$ and $s > n|1/2 - 1/p|$. To prove the endpoint estimate for $s = n|1/2 - 1/p|$, we need following result.

Theorem 4.1. *Suppose that (X, d, μ) is a space of homogeneous type with a dimension n. Suppose that L satisfies ($\mathrm{GGE}_{p_0,p_0',m}$) for some $1 \le p_0 < 2$. Then for every $p \in (p_0, p_0')$, there exists a constant $C = C(n, p) > 0$ such that for all $t \in \mathbb{R}\backslash\{0\}$,*

$$\left\|(I + |t|L)^{-s} e^{itL} f\right\|_p \le C\|f\|_p, \quad s \ge n\Big|\frac{1}{2} - \frac{1}{p}\Big|. \tag{4.1}$$

As a consequence, this estimate (4.1) holds for all $1 < p < \infty$ when the heat kernel of L satisfies a Gaussian upper bound (GE_m).

Proof. We prove this theorem by following the approach in the proof of Theorem 1.1 by using Proposition 2.6 instead of Proposition 2.3. For the details, we leave to the reader. □

Proof of Theorem 1.3. The proof of Theorem 1.3 is inspired by the idea of [39]. Take a function $\Phi \in C^\infty(\mathbb{R})$ such that $\Phi(t) = 0$ if $t < 1/2$ and $\Phi(t) = 1$ if $t > 1$. Define function F by

$$F(u) = I_s(1)(u) - C_s \Phi(u) u^{-s} e^{-iu},$$

where C_s is defined by

$$s \int_{-\infty}^{1} (1 - \lambda)^{s-1} e^{i\lambda u} d\lambda = C_s u^{-s} e^{iu}, \quad u > 0.$$

It is seen that for $0 < u \le 1$ and $k \in \mathbb{N}$,

$$\frac{d^k}{du^k} F(u) \le C,$$

and for $u > 1$ and $k \in \mathbb{N}$,

$$\frac{d^k}{du^k} F(u) \le C u^{-k}.$$

See [39, Lemma 2.1]. Hence, for every $\beta > (n+1)/2$ we have that $\sup_{R>0} \|\eta \delta_R F\|_{C^\beta} \le C$, and so $\sup_{R>0} \|\eta \delta_R F(t\cdot)\|_{C^\beta} \le C$ with a constant $C > 0$ independent of $t > 0$. Then we apply (a) of Proposition 2.7 to know that $F(tL)$ is bounded on $L^p(X)$ for all $p_0 < p < p_0'$. Notice that for every $t > 0$,

$$F(tL) = I_s(t)(L) - C_s \Phi(tL)(tL)^{-s} e^{-itL}. \tag{4.2}$$

This yields that for every $t > 0$,

$$\begin{aligned} \|I_s(t)(L)\|_{p\to p} &\le \|F(tL)\|_{p\to p} + C\|\Phi(tL)(tL)^{-s} e^{-itL}\|_{p\to p} \\ &\le C + C\|\Phi(tL)(tL)^{-s}(1+tL)^s\|_{p\to p}\|(1+tL)^{-s} e^{-itL}\|_{p\to p}. \end{aligned} \tag{4.3}$$

Applying (a) of Proposition 2.7 again, we have that $\|\Phi(tL)(tL)^{-s}(1+tL)^s\|_{p\to p} \le C$. This, in combination with (4.1) in Theorem 4.1, implies $\|I_s(t)(L)\|_{p\to p} \le C$ for $t > 0$.

Since $I_s(t)(L) = \overline{I}_s(-t)(L)$ for $t < 0$, we have that $\|I_s(t)(L)\|_{p\to p} \le C$ for $t < 0$. The proof of Theorem 1.3 is complete. □

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Peng Chen, Department of Mathematics, Sun Yat-sen University, Guangzhou, 510275, P.R. China
E-mail address: chenpeng3@mail.sysu.edu.cn

Xuan Thinh Duong, Department of Mathematics, Macquarie University, NSW 2109, Australia
E-mail address: xuan.duong@mq.edu.au

Ji Li, Department of Mathematics, Macquarie University, NSW, 2109, Australia
E-mail address: ji.li@mq.edu.au

Lixin Yan, Department of Mathematics, Sun Yat-sen (Zhongshan) University, Guangzhou, 510275, P.R. China
E-mail address: mcsylx@mail.sysu.edu.cn